

Lava — Bug Fix Audit Trail

Revision §11.4.44 (2026-06-24, updated again): appended three Crashlytics-triage fixes from 2026-06-24 session — ALL THREE FIXES LANDED in the working tree (the earlier "P1 OWED" note was a stale-git diff race artifact, corrected here): P0 CredentialsKeyHolder locked FATAL (58a1335272bc, fix in CredentialsEntryRepositoryImpl.observe() runCatching, §6.O closure log), P1 LazyColumn 2nd-site nested-scroll FATAL (c7c8cccad09f, fix feature/search_input/.../ SearchInputScreen.kt:96 modifier = Modifier.weight(1f) + §6.Q scanner CHECK 2 — scanner FAIL→PASS, "no nested-scroll antipattern detected"), P2 TopicPageDto MissingFieldException NON_FATAL (8cde0ac208b3, **PARTIAL** — commentsPage default only; the real IA crawl-topic payload omits all 6 fields so the full fix is a tracked follow-up, §6.O closure log). Source: docs/issues/2026-06-24-crashlytics-full-triage.md. All three are reproduce-first + Bluff-Audit'd and ship in 1.3.11-1073.

Revision §11.4.44 (2026-06-24, updated): appended search cancel/timeout fix covering all 3 root causes — SearchResultViewModel back-press cancellation + 25 s client-side withTimeout (Bug 1+2, commit 20d98914) + engine 18 s total-search deadline in handlers/v1/search.go + stale YTS mirror refresh (Bug 3, commits 0e81730b + 1aa42536 + 3c1fa159). 3 client regression tests + 2 engine Go tests added; all falsifiable. Final SHAs confirmed from git show --stat.

Revision §11.4.44 (2026-06-23): appended five fixes from the 2026-06-23 session — H1 search-401 (AuthInterceptor handoff-key overwrite), api-app-17 stale-binary (§6.Z wrong-binary saga), ApiHttpException type regression, ApiKeyClientTest compile break, and firebase-distribute SIGPIPE. Each entry cites its real commit SHA verified against git show --stat.

Revision §11.4.44 (2026-06-08): appended the 2026-06-08 session's six real defect fixes — the four §6.E capability-honesty bluffs (kinozal / nnmclub / rutor magnet wire-through + gutenberG TORRENT_DOWNLOAD drop), the HelixQA broken-pin go.mod conflict that broke lava-api-go's build, and the nav-compose 2.9.0 → 2.9.1 test-teardown lifecycle race. Each entry below cites its real commit SHA + real files (verified against git show --stat). Per §11.4.6 no-guessing, details that could not be confirmed from git are marked UNCONFIRMED.

Per constitutional clause **§6.T.4 (Bugfix Documentation)** — every bug fix in this project **MUST** be documented here with root cause analysis, affected files, fix description, link to the verification test/ challenge, and the commit SHA that landed the fix.

§6.O (Crashlytics-Resolved Issue Coverage Mandate) extends this for Crashlytics-recorded issues; their closure logs live at `.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md`. §6.T.4 covers the rest — operator-reported, self-discovered, or reviewer-flagged bugs that don't enter the Crashlytics pipeline.

Format per entry:

`## YYYY-MM-DD — <short slug>`

`**Root cause:** ...`

`**Affected files:** ...`

`**Fix:** ...`

`**Verification test/challenge:** path or commit ref`

`**Fix commit:** SHA`

`**Forensic anchor:** (optional) what surfaced the bug`

2026-06-16 — 3 circuit-breaker providers: transient timeout tripped the breaker (kinozal, nnmclub, rutracker)

Root cause: kinozal, nnmclub, and rutracker wrap their HTTP calls in a circuit breaker (`c.breaker.Execute(closure)`). The closure had no retry, so a single transient timeout/5xx returned an error to the breaker → counted as a consecutive failure. ~5 transients would OPEN the breaker (~10 s lockout) — strictly WORSE than the plain single-attempt gap, because one slow upstream response could lock out the user for 10 s.

Affected files: `internal/kinozal/client.go` (`Fetch/FetchWithHeaders/PostForm`), `internal/nnmclub/client.go` (4 methods), `internal/rutracker/client.go` (5 methods)

- each provider's `*_test.go` (nnmclub's `client_test.go` is new). Added `maxAttempts=3/retryBackoff=500ms` + a `doWithRetry` helper used INSIDE each breaker `Execute` closure.

Fix: retry transient failures (network/timeout or 5xx) INSIDE the breaker closure, so a recovered transient returns `nil` to the breaker (failure counter untouched, breaker stays Closed). Genuine post-retry failures still return one error to the breaker — real-outage counting is unchanged. Terminal errors (404/403/decode) never retried.

Verification test/challenge: per provider, `TestRetriesOnTransient5xx` (503-once-then-200 → recovers + asserts `breaker.GetFailures()==0 + GetState()==StateClosed` — the load-bearing proof the transient did NOT trip the breaker) + `TestTerminalErrorNotRetried` (404 → 1 attempt, surfaced immediately). Falsifiability rehearsed all 3: bypassing the retry → `TestRetriesOnTransient5xx` FAILS ("expected transient 503 to be retried into success, got err: upstream 503"); reverted → ok. `gofmt` + `go vet` clean.

Fix commit: (this commit) — completes provider retry-resilience for all 10 HTTP providers (this 3 + the 7 below). Same undistributed `api-go` cycle (2.3.31). **Forensic anchor:** nezha audit (3-subagent fan-out) of every provider HTTP client.

2026-06-16 — 6 sibling providers shared the tokyotosho single-attempt-no-retry gap (knaben, nyaa, bitsearch, torrentdownloads, gutenber, flaresolverr)

Root cause: an audit of every provider HTTP client (triggered by the tokyotosho fix below) found the SAME single-attempt-no-retry class in 6 more providers: each issued one `http.Do` with no retry, so a transient upstream timeout/5xx surfaced a user-facing error even though the next request would succeed. Cited from source: `knaben/client.go` (single `Do`), `nyaa/client.go`, `bitsearch/client.go`, `torrentdownloads/client.go`, `internal/gutenberg/client.go` (`getJSON` + `downloadBytes`), `internal/provider/flaresolverr/client.go` (`Get`).

Affected files: the `client.go` of each of the 6 providers (refactored the inline fetch into a bounded-retry loop + a single-attempt helper classifying transient = network/timeout-from-http.Do OR 5xx; terminal = 404/403/401/decode never retried) + each provider's `*_test.go` (added retry + terminal-not-retried tests). `flaresolverr` uses `maxAttempts=2` (its 90s solve budget is already long); `gutenberg` retries BOTH `getJSON` + `downloadBytes` via a shared helper.

Fix: same bounded-retry pattern as `tokyotosho` (`maxAttempts=3` / 500ms backoff, ctx-bounded), adapted to each provider's response shape (RSS/JSON/HTML/latin-1).

Verification test/challenge: per provider, `TestSearch_RetriesOnTransient5xx` (+ download variant for `gutenberg`) and `TestSearch_TerminalErrorNotRetried`. Falsifiability rehearsed for ALL 6: bypassing the retry loop makes the retry test FAIL with the provider's exact HTTP 503: ... unknown error (e.g. `knaben: HTTP 503: provider: unknown error`); reverted → ok. `go vet` + `gofmt` clean.

Fix commit: (this commit) **Forensic anchor:** `nezha` real-tracker E2E surfaced `tokyotosho`; the systemic audit (3-subagent fan-out) surfaced the 6 siblings. Same `api-go` cycle (2.3.31).

2026-06-16 — tokyotosho curated provider: user-facing "unknown error" on slow upstream (no retry on transient failure)

Root cause: `internal/provider/curated/tokyotosho` issued a single HTTP request bounded by `DefaultTimeout = 20s`. Tokyo Toshokan's live latency is variable (measured on the `nezha` heavy-test node 2026-06-16: typical 4–8 s, occasionally >20 s). When a single request exceeded 20 s the client returned `provider: ErrUnknown` ("unknown error") to the user — even though the very next request would have succeeded. Surfaced by the `realtrackers` live test `TestLive_SearchReturnsRealMagnets`, which PASSED at 16:14 (16.5 s) and FAILED at 16:15 (the >20 s case), confirming external-latency intermittency, not a parser bug (direct `curl` probes all returned HTTP 200 with valid RSS).

Affected files: lava-api-go/internal/provider/curated/tokyotosho/client.go (added maxAttempts/retryBackoff; refactored the inline fetch into fetchFeed (bounded-retry loop) + fetchFeedOnce (single attempt classifying transient vs terminal)). Terminal errors (404/403/401/decode) are NOT retried.

Fix: retry transient failures (network/timeout or 5xx) up to 3 attempts with a 500 ms backoff, bounded by the caller's ctx. Converts transient upstream slowness into a successful search while leaving the user-visible result shape and the production DefaultTimeout unchanged.

Verification test/challenge:

tokyotosho_test.go::TestSearch_Retrieves0nTransient5xx (reproduce-first §6.T.1: 503-once-then-fixture → recovers, 2 results) + TestSearch_TerminalErrorNotRetried (404 → ErrNotFound after exactly 1 attempt). Falsifiability rehearsed: bypassing the retry loop → TestSearch_Retrieves0nTransient5xx FAILS with "should recover via retry: tokyotosho: HTTP 503: provider: unknown error"; reverted → PASS.

Fix commit: (this commit) **Forensic anchor:** nezha.local heavy-testing enablement, real-tracker E2E run; operator chose "add bounded retry". Evidence: .lava-ci-evidence/nezha/2026-06-16-system-boot.md.

2026-06-14 — search "Something went wrong" across providers via the on-device API (ApiBackedTrackerClient used the strict client + no per-endpoint key)

Root cause: the dynamic ApiBackedTrackerClient (the client that issues GET | POST /v1/{provider}/{op} against the chosen lava-api-go / on-device api-app) was wired in core/tracker/client/.../di/TrackerClientModule.kt:275/302 with the **unqualified okHttpClient** — which (per NetworkModule.kt:101) uses the system trust store only and is explicitly documented "MUST NOT be used for LAN endpoints." The on-device api-app serves over HTTPS with a **self-signed LAN cert**, so every search failed at the TLS handshake (CertPathValidatorException); and even past TLS, ApiBackedTrackerClient attached **no per-endpoint Lava-Auth key**, so the authenticated /v1/{provider}/search returned **HTTP 401**. Either way getString threw → SearchResultViewModel mapped the Throwable to error_something_goes_wrong ("Something went wrong, please try again"). This is the exact SP-3.1 / Defect-A class that was fixed for the /providers catalogue fetch (ProviderCatalogRepository got the @Named("lan") client + goApi.key) but **never fixed for the per-provider search/browse/topic/download/login path**.

Affected files: core/tracker/client/.../di/TrackerClientModule.kt (factory → @Named("lan") client + @Named("authFieldName") + authKey = ApiBaseUrlHolder.currentKey()); core/tracker/client/.../ApiBackedTrackerClient.kt (+authFieldName/authKey ctor params +

Request.Builder.withAuth() on every request); core/tracker/client/.../
ApiBaseUrlHolder.kt (now holds the per-endpoint key); core/domain/.../
RepopulateProvidersOnStartupUseCase.kt

- ActiveApiBaseUrlActivator (cold-start threads the key); feature/
onboarding/.../OnboardingViewModel.kt
(ApiBaseUrlHolder.set(apiBaseUrl, goApi.key)); app/.../
StartupProvidersModule.kt.

Fix: the dynamic client now uses the permissive-TLS @Named("lan") client (accepts the self-signed LAN cert) and attaches the active endpoint's per-instance key as the Lava-Auth header on EVERY /v1 request — mirroring the proven Defect-A pattern. The key is threaded from onboarding AND the cold-start re-populate.

Verification test/challenge:

ApiBackedTrackerClientTest.search_attachesPerEndpointAuthKey_soAuthGatedApiReturnsResults (real client + MockWebServer that 401s without the key → asserts a real result row

- the Lava-Auth: k header on the wire) + ...
search_withoutAuthKey_throwsOnAuthGatedApi (discriminator) +
RepopulateProvidersOnStartupUseCaseTest (asserts the cold-start activator receives the key). Falsifiability (re-performed): neuter
ApiBackedTrackerClient.withAuth() → the search test FAILS with HTTP 401; reverted → green. Device-level proof (search on the VM against the real api-app) is the final gate via the §6.AE/HelixQA video pass.

Fix commit: (this commit).

Forensic anchor: operator report (2026-06-14) — "onboarded RuTracker, YTS, Kinozal; searched 'prince' via the Android API with all 3 selected → 'Error, Something went wrong, please try again'."

Update (2026-06-14) — L5 root cause FOUND + fixed (api-app ApiKeyProvider), release scope now client + api-app, status: on-device verification in progress — NOT yet shipped. After the client-side fixes above (now framed as L1–L4 of a 5-layer cascade — see docs/issues/2026-06-14-search-multilayer-rootcause.md), search STILL returned 401 on a clean matched pair (granted=true, engine up). The load-bearing fifth layer (L5) was an **api-app** defect, not a client one: api-app/src/main/kotlin/lava/api/app/handoff/ApiKeyProvider.kt cached its key and port resolver lambdas in ContentProvider.onCreate(), gated on the ApiApplication.controllerHolder/keyStoreHolder being non-null, on the FALSE inline comment "ContentProvider.onCreate runs AFTER Application.onCreate." Android runs ContentProvider.onCreate **BEFORE** Application.onCreate, and the holders are set inside ApiApplication.onCreate (lines 46-49) — so at cache time the holders were **always null**, the resolver lambdas stayed { null } for the whole process, and the provider served an **empty cursor**. The full key-loss chain: client ApiKeyClient.read() → null → withLocalApiKeyIfMissing → keyless endpoint → ApiBaseUrlHolder.set(url, null) → ApiBackedTrackerClient built with authKey=null → no Lava-Auth header → every /v1/{provider}/search 401. Public routes (/providers, /health) need no key, so they worked — which is why search NEVER worked while everything else did. The existing

withFakes unit test was a **bluff**: it injected the resolver lambdas through a test-only seam, bypassing the real onCreate → holder path where the bug lives, so it passed green against the empty-cursor production path. **Fix (applied)**: ApiKeyProvider resolves the holders + the engine Running state **LAZILY per query()** (resolveRunningKey() / resolveRunningPort() defaults), removing the onCreate-ordering dependency. **On-device pinpoint**: .lava-ci-evidence/search-verification/2026-06-14-keyloss-pinpoint.md (LAVAKEYDBG logcat with LocalApiKey readerSet=true readLen=-1 = key read returned null at the source). **Verification OWED**: a real-holder regression test driving the actual onCreate ordering (no withFakes seam) so the empty-cursor defect fails RED before the fix. **Release scope change**: 1.3.9 is NO LONGER client-only — the L5 fix is in the api-app, so **client 1.3.9-1066 + api-app 0.2.9-14 ship together** (§6.Y bump applied). **Status: fix applied, on-device verification in progress — NOT yet shipped**. Search is NOT claimed to work on-device until the device gate (authed / v1/{provider}/search → 200 + result rows) is GREEN.

2026-06-13 — curated TPB + Torrents-CSV providers single-domain rotation risk (no mirror failover)

Root cause: the embedded curated providers thepiratebay (apibay.org) and torrentscsv (torrents-csv.com) each hardcoded a SINGLE base domain in their Client. Public-tracker domains rotate frequently under takedown pressure — the companion YTS provider was caught by exactly this on 2026-06-13 (yts.mx went NXDOMAIN on public DNS while yts.bz still served HTTP 200). A single-domain client goes silently unreachable the moment that one domain drops, so the provider would appear in the on-device GET /providers catalogue but every search would error for the user — a reachability bluff waiting to happen.

Fix: refactored both clients to the YTS mirror-failover pattern — NewClientWithMirrors(DefaultBaseURLs), a first-live-mirror-wins loop in Search, and a perAttemptTimeout (8s) cap per mirror so one dead/hanging host cannot stall the whole search. New() registration now goes through the mirror list. The failover ARCHITECTURE is the load-bearing change; the production DefaultBaseURLs for each provider deliberately contains only ONE host because live HTTP probing on 2026-06-13 found exactly one real endpoint per provider:

- **TPB**: apibay.org/q.php = HTTP 200 JSON; the TPB frontend proxies (thepiratebay.org/apibay 302→SPA, tpb.party / thepiratebay10.xyz 404 /q.php) expose NO compatible JSON API. apibay.org is TPB's canonical internal API.
- **Torrents-CSV**: torrents-csv.com/service/search = HTTP 200 JSON; torrents-csv.ml = 404, torrents-csv.org / api.torrents-csv.com = NXDOMAIN. torrents-csv.com is the canonical public instance (git.torrents-csv.com is the Gitea code repo, not a search API). Per §6.L, fabricating dead .xyz/.party/.ml/.org mirror entries would be a bluff (they'd only add latency + failover noise), so each list stays a real single-element slice; a future real mirror is a one-line addition with no client change. YTS itself genuinely has 4 live mirrors and keeps its multi-host list.

Affected files: lava-api-go/internal/provider/curated/thepiratebay/client.go, .../thepiratebay/provider.go, .../thepiratebay/thepiratebay_test.go, .../torrentscsv/client.go, .../torrentscsv/provider.go, .../torrentscsv/torrentscsv_test.go, core/apiengine/src/main/resources/api-source.hash (recomputed for the embed).

Verification test/challenge: new fast fixture (httptest, no live net) tests in each package: TestSearch_FailsOverToHealthyMirror (dead 503 first mirror → returns the SECOND healthy mirror's real parsed SearchItem rows — primary assertion on user-visible info_hash, not a call count) + TestSearch_AllMirrorsDownSurfacesError (all mirrors error → no fake empty success). go test ./internal/provider/curated/... = all ok.

Bluff-Audit: TPB TestSearch_FailsOverToHealthyMirror — Mutation: add return nil, lastErr after the first loop iteration in Search (break failover after first mirror). Observed: Search across [dead, healthy] mirrors should fail over, got error: thepiratebay: HTTP 503: provider: unknown error. Torrents-CSV TestSearch_FailsOverToHealthyMirror — same mutation. Observed: Search across [dead, healthy] mirrors should fail over, got error: torrentscsv: HTTP 503: provider: unknown error. Reverted: yes (both re-run ok).

Fix commit: (this worktree commit — see git log)

Forensic anchor: YTS domain-failover fix .lava-ci-evidence/bluff-hunt/2026-06-13-yts-domain-failover.json; the same single-domain rotation class applied to the two sibling curated providers.

2026-06-13 — Tokyo Toshokan curated provider added (Defect B — anime/Asian-media RSS, honest CapSearch)

Type: feature (curated embedded provider), logged here per §6.T.4 because it extends the user-reachable provider catalogue.

What: new compiled-in curated provider tokyotosho (lava-api-go/internal/provider/curated/tokyotosho/) sourcing the public Tokyo Toshokan RSS search feed (<https://www.tokyotosho.info/rss.php?terms=<q>>), AuthType NONE / SupportsAnonymous true, capabilities SEARCH + MAGNET_LINK. Registered in curated.RegisterAll (both startup paths). Broadens the anime/Asian-media niche alongside nyaa.

Wire-shape specifics (differ from nyaa): the magnet is embedded in each item's <description> CDATA HTML (<a href="magnet:?xt=urn:btih:<HASH>...">), extracted by regex; the info_hash is BASE32 (32 chars), not 40-hex; the RSS carries NO seeders field, so Seeders defaults to 0 (documented, honest absence); size is decimal SI (113.89MB → 113890000 bytes), not nyaa's IEC.

§6.E honesty (the load-bearing check): the terms= parameter genuinely filters — verified LIVE 2026-06-13 (real HTTP): terms=naruto → 93% of titles contain "naruto"; terms=bleach → 98% contain "bleach"; the empty feed contains neither; naruto vs bleach result sets are near-disjoint (2 of ~140 overlap). So CapSearch is

honest — NOT the EZTV/Knaben-search_type no-op-query bluff. The `// go:build realtrackers TestLive_QueryActuallyFilters` encodes both proofs (term-match-rate $\geq 50\%$ + $\leq 50\%$ cross-query info_hash overlap) and PASSED against the real endpoint.

Affected files: `lava-api-go/internal/provider/curated/tokyotosho/{client.go,provider.go,tokyotosho_test.go, live_realtrackers_test.go, testdata/tokyotosho_naruto.xml}` (new); `lava-api-go/internal/provider/curated/curated.go` (one import + one `r.Register` line); `core/apiengine/src/main/resources/api-source.hash` (recomputed).

Verification: fixture test `tokyotosho_test.go` (real Client + `httptest.Server` serving captured real RSS bytes; asserts parsed title/infohash/magnet/size/no-magnet-filter)

- live `live_realtrackers_test.go` (`TestLive_SearchReturnsRealMagnets` + `TestLive_QueryActuallyFilters`, both PASS against the real endpoint).
`go test ./internal/provider/curated/... = all ok.`

Bluff-Audit (§6.J clause 2 / Seventh Law clause 1):

- Mutation: `extractInfoHash` returns "" unconditionally (magnet extraction off) → `TestSearch_ParsesFixture` FAILED got 0 results, want 2 (the no-magnet row must be filtered). Reverted.
- Mutation: send the query as xterms not terms (query-not-sent under the honored name) → `TestSearch_SendsTermsParam` FAILED Search: tokyotosho: HTTP 400: provider: unknown error. Reverted.
- Mutation: MB multiplier `1e6` → `1<20` (IEC instead of decimal SI) → `TestSearch_ParsesFixture` FAILED `sizeBytes = 119422320`, want `113890000` (113.89 MB decimal). Reverted.

Fix commit: see this commit (worktree).

2026-06-13 — onboarding catalogue fetch 401 against a freshly-discovered API (GET /providers auth-gated)

Root cause: `lava-api-go/internal/router/router.go` registered `engine.GET("/providers", ...)` AFTER `engine.Use(auth.GinMiddleware())` (and the `auth.NewMiddleware` chain). The provider catalogue is the endpoint the Android client fetches during ONBOARDING against an API it has just discovered on the LAN — at which point the client holds no pre-shared Lava-Auth key with that API. The auth gate therefore rejected the fetch with HTTP 401 → `OnboardingViewModel fetchAndPopulateProviders` caught the failure and surfaced `PROVIDER_CATALOG_FALLBACK_NOTICE` ("Couldn't reach the selected API — showing bundled providers"), so a perfectly reachable API showed the bundled fallback. This is distinct from Defect A (2026-06-12, a TLS+missing-key client problem on the *client* side); Defect A's fix made the client send the key, but the api-app's allowlist need not include a just-discovered client at all — the catalogue is public metadata and must not be gated in the first place.

Affected files: lava-api-go/internal/router/router.go (registration moved before the auth middleware, with rationale comment + a NOTE where it used to live); lava-api-go/internal/router/router_config_wiring_test.go (+TestBuild_ProvidersOpenWithFullAuthChain); lava-api-go/tests/contract/providers_public_auth_boundary_test.go (new, real-binary 3-way boundary); feature/onboarding/src/test/kotlin/lava/onboarding/OnboardingViewModelDynamicProvidersTest.kt (+replace-not-merge success test + no-banner-on-success assertion); core/apiengine/src/main/resources/api-source.hash (recomputed).

Fix: register GET /providers BEFORE the auth middleware, alongside /health+/ready. It is public, non-sensitive provider metadata (ids, capabilities, authType, baseUrls — no credentials, no user data). Per-provider operations under /v1/:provider/... remain fully auth-gated. The standalone binary and the embedded api-app share the exact router.Build, so the one edit fixes both surfaces (DRY; a divergent embed router would be a §6.J bluff vector).

Verification test/challenge: server

TestBuild_ProvidersOpenWithFullAuthChain (unit) + TestProviders_PublicCatalogue_PerProviderStillGated (real-binary e2e, proves unauth /providers→200 while unauth /v1/...→401 and authed /v1/... crosses the gate); client OnboardingViewModelDynamicProvidersTest (success → exactly the API's set, bundled-absent, no banner; failure → banner + non-blank). Every layer's falsifiability mutation re-performed by the main stream (not trusted from the subagent worktree): gate /providers → e2e sub-test A 401; drop populateFrom → replace test FAILED; success-emits-banner → assertNull FAILED. All reverted GREEN.

Fix commit: 9ae9ab90 (server fix) + 132d1b07 (real-binary e2e) + 99e5893a/6ce100bc (client full-flow tests).

Forensic anchor: operator real-device report (2026-06-13) — "choose discovered API 192.168.0.107:8443 → next screen shows 'Couldn't reach the selected API — showing bundled providers'". Crashlytics non-fatal 47b000d5 (provider_catalog_fetch_failed, HTTP 401).

2026-06-12 — provider-catalogue fetch fails self-signed-LAN TLS handshake (Defect A — "only 4 providers")

Root cause: ProviderCatalogRepository injected the *unqualified* strict-TLS OkHttpClient (system trust store only) and sent no auth key. The on-device api-app serves GET /providers over a self-signed LAN cert behind the Lava-Auth gate, so every real-device fetch died at the TLS handshake (CertPathValidatorException: Trust anchor ... not found) → Result.failure → onboarding fell back to the 4 bundled verified && apiSupported descriptors. **Real-device proof:** Crashlytics NON_FATAL 042b9b61 (provider_catalog_fetch_failed), 1.3.3-1060, Galaxy S23 Ultra / Android 16. **Affected files:** core/data/.../ProviderCatalogRepository.kt (inject @Named("lan") client + @Named("authFieldName"); fetchProviders(apiBaseUrl, authKey) + withAuthKey), core/domain/.../FetchProvidersUseCase.kt (thread authKey), feature/onboarding/.../

OnboardingViewModel.kt (pass Endpoint.GoApi.key), core/data/build.gradle.kts + gradle/libs.versions.toml (okhttp-tls for self-signed-HTTPS tests). **Fix:** route the catalogue fetch through the permissive LAN client (accepts the self-signed cert) + attach the endpoint's per-instance key — mirrors the existing NetworkApiRepositoryImpl.getApi() Endpoint.GoApi path. **Verification test/challenge:** ProviderCatalogRepositoryTest rewritten to cross a real self-signed-HTTPS + auth-gated MockWebServer (6/6; strictClientFailsTheHandshake = codified pre-fix mutation; missingAuthKeyIsRejectedByTheGate proves the header is load-bearing); OnboardingViewModelDynamicProvidersTest green. **Fix commit:** 0deb54e7 **Forensic anchor:** operator "only 4 providers in onboarding wizard"; §6.O closure log .lava-ci-evidence/crashlytics-resolved/2026-06-12-provider-catalog-fetch-tls.md. **Scope note:** Defect B (embedded Jactett in the api-app) still OPEN — the client now shows whatever the api-app serves, but the api-app serves only native providers until embedded Jactett lands.

2026-06-02 — api-app-ondevice-challenges-three-defects + harness-isolation

The Containers emulator-matrix CLI gained a generic --gradle-module flag (Containers commit 9a61a153); the Lava glue scripts/run-api-app-challenge-matrix.sh now forwards it so the :api-app Compose UI Challenges (C01–C04) finally **execute** against the :api-app module instead of a 0-test false-green against :app. Running them for real surfaced **three latent product defects** (all previously invisible because the tests had never actually run) plus a test-harness isolation flaw. All proven fixed: clean sequential 4/4 green on a cold-booted Pixel_8/API35 via the Containers runner (host-direct+HVF, gating=true).

Root cause 1 (C02 — mDNS service-name shadow → API35 crash): in NsdMdnsAdvertiser.register(), NsdServiceInfo().apply { this.serviceName = serviceName } — inside apply{} the implicit receiver is the NsdServiceInfo, whose own serviceName property **shadowed** the advertiser's constructor val, so the RHS serviceName resolved to the receiver's (null) value. The service registered with an EMPTY name → Android 15 NsdManager.validateService threw IllegalArgumentException ("The service name or the service type is missing") → uncaught → engine start crashed → instrumentation process crash (C01 ok, C02 crashed, C03/C04 never ran).

Root cause 2 (C03 — cross-test native-engine pollution): the Go embed's internal/mobile.current is process-global, but Hilt rebuilds the @Singleton ApiEngineController per test. C02 starts the engine and never stops it; C03 then tries to Start → mobile.Start sees current != nil → "already running" → C03's start→Running wait times out.

Root cause 3 (C04 — notification restart-after-stop): two bugs. (a) ApiEngineController.restart() called stop() first unconditionally; after a Stop the engine is already stopped → mobile.Stop returns "no server running" → Error state → restart() bailed before start(). (b) ApiEngineService's state collector called stopSelf() on ANY Stopped, including the INITIAL Stopped a freshly-recreated Service sees while handling an ACTION_RESTART — destroying the Service and cancelling the restart coroutine before it could drive the engine up.

Root cause 4 (harness isolation flakiness): the foreground `ApiEngineService` is a process-singleton that outlives the per-test `Hilt @Singleton` controller, so a stale `Service` from a prior test intercepted the next test's start/notification actions (driving the wrong, stale controller) — producing run-to-run inconsistent failures. Plus the on-device `OnDeviceApiClient` 15s `readTimeout` flaked the HTTP/2 stream (`SocketTimeoutException`) on the emulator's first cold-serve.

Affected files:

- `submodules/containers/.../cmd/emulator-matrix + pkg/emulator/*` (the `--gradle-module` flag, Containers commit 9a61a153)
- `scripts/run-api-app-challenge-matrix.sh` (forward `--gradle-module`)
- `api-app/src/main/kotlin/lava/api/app/service/NsdMdnsAdvertiser.kt` (RC1)
- `api-app/src/main/kotlin/lava/api/app/control/ApiEngineController.kt` (RC3a)
- `api-app/src/main/kotlin/lava/api/app/service/ApiEngineService.kt` (RC3b)
- `api-app/src/androidTest/.../Challenge0{1,2,3,4}*Test.kt` (RC2 + RC4 @Before reset + Service teardown)
- `api-app/src/androidTest/.../OnDeviceApiClient.kt` (RC4 timeout)
- `lava-api-go/internal/mobile/restart_repro_test.go` (NEW Go same-port-restart coverage — proved the Go embed restart is innocent)

Fix:

1. RC1 — capture `serviceName/serviceType` into `LOCALS` before the `apply{}` (kills the shadow) + wrap `registerService` in `runCatching` so a future platform rejection degrades gracefully instead of crashing the engine.
2. RC2 + RC4 — each `Challenge @Before` resets the process-global native engine (`NativeApiEngine().stop()`) AND kills any stale foreground `Service` (`stopService`).
3. RC3a — `restart()` skips `stop()` when already `Stopped` (just `start()`).
4. RC3b — the `Service` collector self-stops only on a genuine `running→stopped` transition (`sawRunning` flag), never on the initial `Stopped`.
5. RC4 — `OnDeviceApiClient` connect/read timeouts 10/15s → 20/30s.

Verification test/challenge: `:api-app Challenge C01–C04 EXECUTED` green on cold-booted Pixel_8/API35 via the Containers runner (`gating=true`, `host-direct+HVF`). Evidence: `.lava-ci-evidence/phase-e-api-app/2026-06-02T11-32-48Z-gate/` (+ a confirmation run). Each fix falsifiability-rehearsed; the gate itself is the load-bearing on-device acceptance gate (§6.AE / §6.Z). **Fix commit:** 0deb54e7 (parent) + Containers 9a61a153. **Forensic anchor:** the `--gradle-module` flag made the never-before-run `:api-app` Challenges execute for real — the textbook §6.J/§6.L payoff: three real product defects had been invisible behind a 0-test false-green.

2026-05-06 — phase1-distribute-three-bugs

The first real-distribute of `Lava-Android-1.2.7-1027` + `lava-api-go-2.1.0` exposed three bugs that landed in commit e947081:

Root cause 1: /health and /ready endpoints were registered AFTER the auth middleware in lava-api-go/cmd/lava-api-go/main.go buildRouter, so the orchestrator's liveness probe got 401 → restart loop.

Root cause 2: core/network/impl/.../LavaAuthBlobProvider.kt was declared internal, but the Phase 11 build-time-generated lava.auth.LavaAuthGenerated class lives in :app's source set — different module → cannot access internal interface → compile failure.

Root cause 3: scripts/distribute-api-remote.sh + deployment/thinker/thinker-up.sh did NOT ship the operator's local .env LAVA_AUTH_* + LAVA_API_HTTP3_*/BROTLI_*/PROTOCOL_* values to the thinker.local container. The new 2.1.0 binary required LAVA_AUTH_FIELD_NAME + LAVA_AUTH_HMAC_SECRET at boot → crash-loop.

Affected files:

- lava-api-go/cmd/lava-api-go/main.go
- core/network/impl/src/main/kotlin/lava/network/impl/LavaAuthBlobProvider.kt
- scripts/distribute-api-remote.sh
- deployment/thinker/thinker-up.sh

Fix:

1. Register /health + /ready BEFORE the auth chain.
2. Drop internal from LavaAuthBlobProvider.
3. Distribute script merges operator's .env auth/transport block into a temp env file before scp; thinker-up iterates the variables and passes each as -e VAR=\$VAR to podman run.

Verification test/challenge: post-fix smoke test in scripts/distribute-api-remote.sh's 60-second /health wait (passes); curl -fsSk https://thinker.local:8443/{health,ready} return {"status":"alive"} / {"status":"ready"} (passes); auth gate returns 401 with {"error":"unauthorized"} for missing header (passes — fail-closed posture confirmed).

Fix commit: e947081

Forensic anchor: the very first bash scripts/distribute-api-remote.sh after the api-go-2.1.0 build showed lava-api-go: config: config: LAVA_AUTH_FIELD_NAME is required in a tight loop, with the orchestrator marking the container (unhealthy) and triggering the restart loop. Operator ran the distribute, got the lock-up output, immediately surfaced the issue.

2026-05-12 — c03-anonymous-toggle-checkauth-bluff

Root cause: OnboardingViewModel.onTestAndContinue() (the "Test & Continue" button handler on the Configure step) called sdk.checkAuth(currentId) for BOTH AuthType.NONE providers AND

`config.useAnonymous=true` paths, then asserted that the result equals `AuthState.Authenticated`. For users opting INTO anonymous mode on a `FORM_LOGIN` tracker (RuTor with the toggle on), `checkAuth` correctly returns `Unauthenticated` — that IS the user's chosen state. The code treated it as a failure (`error = "Connection failed"`) and never advanced to the Summary step. The wait-for-"All set!" timeout fired and the user was stuck.

The C03 Challenge Test detected this only as a 60s timeout — no indication of WHY. The bug was invisible to the Sixth Law clauses 1-5 because:

- The test's primary assertion was on "All set!" appearing (✓ correct per clause 3).
- The production stack was traversed end-to-end (✓ correct per clauses 1 + 4).
- But there was no logged stack trace; the catch block called `analytics.recordNonFatal` (Crashlytics-only) and silently set the error state. The diagnostic gap was the source of the bluff: green test infrastructure + green code review + green Bluff-Audit on prior commits all missed it because nobody had run C03 end-to-end on a device with logcat tailing to see the error.

Affected files:

- `feature/onboarding/src/main/kotlin/lava/onboarding/OnboardingViewModel.kt`

Fix:

1. Skip `sdk.checkAuth(currentId)` entirely on the anonymous branch. Anonymous = user opted out of auth = no auth state to check = not a failure when there's no session.
2. Added diagnostic `logger.d/logger.e` breadcrumbs on the entire `onTestAndContinue` flow (auth path taken, result, exception). This closes the §6.T.4-spirit gap that made the C03 bug invisible — future failures will print stack traces to logcat.
3. Added `HttpTimeout`, `UserAgent`, and `Logging` plugins to the `rutracker Ktor HttpClient` + 30s timeouts on the main `OkHttpClient`. The HTTP improvements did NOT close the C02 issue (Cloudflare-side stall — see §4.5.3b in `CONTINUATION.md`), but they raise the time-budget for slow networks and move the failure mode from `SocketTimeoutException` to a more diagnosable signal.
4. Bumped Challenge02 test's wait for "All set!" from 30s to 90s to align with realistic real-network round-trip times after the timeout fixes.

Verification test/challenge:

- C03 (`Challenge03AnonymousSearchOnRuTorTest`) — PASS on CZ_API34_Phone API 34 (live emulator at localhost:5555), 2026-05-12 09:25:14 to 09:25:23, total ~9.7s. Evidence at `.lava-ci-evidence/Lava-Android-1.2.13-1033/post-mortem/logcat-c03-fix.txt` shows anon path: `switchTracker(rutor) → test ok: advance to next/Summary → Perform Finish`.
- Falsifiability rehearsal: re-introducing the broken `if (result != null && result != AuthState.Authenticated)` check in the anonymous branch causes C03 to time out at 60s with the Configure screen still visible (the pre-fix failure mode). Verified through the diagnostic logcat: `checkAuth(rutor) result=Unauthenticated → error path triggered → no advance`.

Fix commit: *(this commit)*

Forensic anchor: Phase 1 systematic-debugging session 2026-05-12. Operator picked option C ("investigate C02/C03 properly") + invoked the anti-bluff mandate. Matrix runs on CZ_API34_Phone surfaced the 60s timeout; live-emulator instrumentation captured the anon path: `checkAuth(rutor)`
`result=Unauthenticated` line that identified the bug.

2026-05-12 — onboarding-action-logger-leaks-credentials

Root cause: Discovered during the C03 investigation above — `OnboardingViewModel.perform()` logged the action via `logger.d { "Perform $action" }`. For sealed-class subtypes `OnboardingAction.UsernameChanged(value: String)` and `OnboardingAction.PasswordChanged(value: String)`, Kotlin's auto-generated `toString` includes the value, so `logcat` (and any analytics breadcrumb that picks it up) prints the operator's real RuTracker username + password in plain text. Severity: §6.H credential inviolability concern — although `.env` is gitignored and the values never reach disk via a log file in the normal user flow, on a device with `adb logcat` running (or with `logcat` being collected by a 3rd party app, or shared in a bug report) this is a leak surface.

Affected files:

- `feature/onboarding/src/main/kotlin/lava/onboarding/OnboardingViewModel.kt`

Fix: changed the log expression from `logger.d { "Perform $action" }` to `logger.d { "Perform ${action::class.simpleName}" }` — prints only the action type name, never any of its values.

Verification test/challenge: `logcat` tail during a re-run of C02 shows `Perform UsernameChanged (no value)` and `Perform PasswordChanged (no value)`.

Fix commit: *(same commit as the c03-anonymous-toggle-checkauth-bluff entry above)*

Forensic anchor: `logcat` captures during the C02/C03 diag run on 2026-05-12 surfaced the bug. Pre-fix `logcats` are gitignored under `.lava-ci-evidence/**/post-mortem/` to prevent accidental commit of the captured credentials; only matrix attestations + `gradle.logs` + JUnit XML test-reports are committed.

2026-05-12 — rutracker-cloudflare-mitigation + cookie-selection-bug

Root cause 1 (Cloudflare anti-bot stall): `provideRuTrackerHttpClient` in `:core:tracker:client` constructed a `Ktor + OkHttp HttpClient` with no cookies, no `Accept-Language`, no `Accept`, no `Accept-Encoding`, and no explicit User-

Agent — i.e. a client whose HTTP/2 header shape is a flag-raising fingerprint to Cloudflare's anti-bot. A bare POST to `rutracker.org/forum/login.php` was accepted by the TCP+TLS layer but never received a response body — verified via Ktor Logging plugin output. Host-side curl with normal browser-like headers returns HTTP/2 200 in <1s for the same URL.

Root cause 2 (cookie selection picked wrong cookie as session token):

`RuTrackerInnerApiImpl.login()` previously extracted `token = cookies.firstOrNull { !it.contains("bb_ssl") }` from the final-hop Set-Cookie headers. When Cloudflare in front of rutracker started emitting `cf_clearance=...` on every response, this filter started selecting `cf_clearance` as the "rutracker session token". The subsequent `mainPage(token)` GET then sent `Cookie: cf_clearance=...` without the actual `bb_data` session token, so rutracker returned the guest page and `parseUserId` couldn't find the logged-in element.

Affected files:

- `core/tracker/client/src/main/kotlin/lava/tracker/client/di/TrackerClientModule.kt`
- `core/tracker/rutracker/src/main/kotlin/lava/tracker/rutracker/impl/RuTrackerInnerApiImpl.kt`

Fix:

1. Installed `HttpCookies` plugin (`AcceptAllCookiesStorage`) on the `rutracker HttpClient`.
2. Added browser-class `defaultRequest` headers (`Accept`, `Accept-Language`, `Accept-Encoding`) + `Chrome 124 / Android 14 / Pixel 8 User-Agent`.
3. Added a `runCatching { httpClient.get(Index).bodyAsText() }` pre-flight inside `RuTrackerInnerApiImpl.login()` so the `cf_clearance` cookie lands in the cookie jar before the POST.
4. Tightened the post-login token extraction to match by name prefix (`bb_data / bb_session / bb_login`) instead of the fragile `!contains("bb_ssl")` negation.

Verification test/challenge: Ktor wire-log on C02 re-run shows

- 10:16:58 GET `/forum/index.php` → 200 (~1s) — pre-flight succeeds, cookies stored
- 10:16:59 POST `/forum/login.php` → 302 (was: 60s timeout)
- 10:16:59 GET `/forum/index.php` (auto-redirect) → 200 (~1.3s) — logged-in page returned

The Cloudflare stall is fully resolved by this commit. **C02 itself does NOT yet pass end-to-end** because `GetCurrentProfileUseCase.parseUserId` throws on the post-login page — `#logged-in-username` element isn't in the served HTML (selector stale or mobile-shaped variant). That's a separate domain-archaeology task documented in `docs/CONTINUATION.md §4.5.3c`.

Fix commit: (*this commit*)

Forensic anchor: continued Phase 1 systematic-debugging session 2026-05-12 after operator's "continue" directive. C03 fix landed in 4d27c07; the same session investigated whether the residual C02 failure was Lava-fixable. CF mitigation succeeded; the remaining parser failure surfaced a separate bug that's deferred to a follow-up SP for rutracker HTML parser refresh.

2026-05-12 — c16-stale-assumption-bluff + parser-selector-fallback + firebase-test-improvement

These three changes are landed together as the anti-bluff-audit response to the operator's SIXTEENTH §6.L invocation on 2026-05-12.

Root cause 1 (Challenge16 stale-assumption bluff):

Challenge16ApiSupportedFilterTest asserted "Internet Archive must NOT appear in the onboarding provider list" because at the time of writing (Phase 12 α -hotfix) ArchiveOrgDescriptor.apiSupported was false. Phase 2b (Lava-API per-provider routing) later flipped that to true without updating the test. The test continued to pass green because its waitUntil clause accepted the Welcome screen — where no provider list renders — making "Internet Archive absent" trivially true. Textbook §6.L bluff: green-light while the actual behavior was the OPPOSITE of the claim.

Root cause 2 (GetCurrentProfileUseCase.parseUserId brittle selector): The parser used a single Jsoup selector #logged-in-username to extract the rutracker user-id post-login. With the Cloudflare anti-bot mitigation landing in commit f7d0a62, the login POST now reaches a 302→200 chain successfully — but the post-login page rutracker.org serves to the mitigated client doesn't contain #logged-in-username (selector stale or mobile-shaped variant). Single-selector parsers fail catastrophically when the served HTML changes; multi-selector fallback gives graceful degradation.

Root cause 3 (FirebaseAnalyticsTrackerTest verify-only assertions): Two tests in FirebaseAnalyticsTrackerTest used verify { mock.foo() } as the SOLE assertion — Forbidden Test Pattern per §6.L clause 4 ("Verification-only assertions"). Call-counting is permitted as a secondary signal, never the primary one.

Affected files:

- app/src/androidTest/kotlin/lava/app/challenges/Challenge16ApiSupportedFilterTest.kt
- core/tracker/rutracker/src/main/kotlin/lava/tracker/rutracker/domain/GetCurrentProfileUseCase.kt
- app/src/test/kotlin/digital/vasic/lava/client/firebase/FirebaseAnalyticsTrackerTest.kt
- CLAUDE.md, AGENTS.md, lava-api-go/CLAUDE.md (§6.L count + 16th invocation forensic anchor)
- docs/CONTINUATION.md (audit findings + verification log)

Fix:

1. C16 rewritten to navigate to "Pick your providers" and assert that all 4 verified+apiSupported providers (RuTracker.org, RuTor.info, Internet Archive,

Project Gutenberg) actually render in the list. Falsifiability rehearsal documented in the KDoc: setting `archiveorg apiSupported=false` causes the IA assertion to fail.

2. `GetCurrentProfileUseCase` now tries 4 selectors in order before erroring: `#logged-in-username`, `a.logged-in-as-username`, `.menu-userctrl a[href*='profile.php?u=']`, `a[href*='profile.php?u=']`. Error message updated to "logged-in user-id not found — page may be guest, or selectors stale" so future agents have a clearer diagnostic anchor.
3. `FirebaseAnalyticsTrackerTest`'s verify-only tests refactored to use `mockk slot captures with assertEquals` on the captured String values + `assertTrue(slot.isCaptured)` as a non-zero-count gate. The captured-value equality is now the primary assertion, per §6.L clause 4.
4. Constitutional count update in `CLAUDE.md §6.L (THIRTEEN → SIXTEEN)`, `AGENTS.md §6.L (THIRTEEN → SIXTEEN)`, `lava-api-go/CLAUDE.md §6.L (TEN → SIXTEEN)`. Forensic anchor for the 16th invocation appended to the wall.

Verification test/challenge:

- C16 rewritten: PASS on CZ_API34_Phone API 34 live emulator (2026-05-12 10:50, BUILD SUCCESSFUL).
- 14 of 23 Challenge Tests verified PASS on live emulator this session: C00, C01, C03, C11, C12, C13, C14, C15, C16(rewritten), C20, C21, C22 (in isolation; sweep-mode fails due to C21 back-press state leak — not a real-user bug), C23, C24.
- C02 still does not pass end-to-end (`parseUserId` fails even with fallback selectors). The Cloudflare-mitigation portion of C02 is fully verified working.

Fix commit: (*this commit*)

Forensic anchor: operator's SIXTEENTH §6.L invocation, 2026-05-12, verbatim: "Do EVERYTHING NOW!!! ... Make sure that all existing tests and Challenges do work in anti-bluff manner ... they MUST confirm that all tested codebase really works as expected! ... execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

2026-06-02 — apiengine-missing-serialization-plugin

Root cause: `:core:apiengine (core/apiengine/build.gradle.kts)` applied only `id("lava.android.library")` and did NOT apply the `kotlinx-serialization compiler plugin (id("lava.kotlin.serialization"))`. `NativeApiEngine.start()` calls `json.encodeToString(config.toDto())` on the `@Serializable ConfigDto` (and `status()` decodes `@Serializable StatusDto`), but without the compiler plugin no serializers are generated, so the reflective lookup threw at runtime: `kotlinx.serialization.SerializationException: Serializer for class 'ConfigDto' is not found. Please ensure that class is marked as '@Serializable' and that the serialization compiler plugin is applied. This is independent of R8 — the DEBUG build reproduces it. The on-device controller mapped the failure to Error, so the :api-app landing screen never`

reached "Running"; the Phase E real-device Challenges C02/C03/C04 timed out waiting for "Running". Fake-based unit tests (FakeApiEngine) passed because they bypass the real serializer — the canonical §6.J/§6.Z bluff class.

Affected files:

- `core/apiengine/build.gradle.kts` — applied `id("lava.kotlin.serialization")`; removed the now-redundant explicit `kotlinx.serialization.json dep` (the convention plugin contributes it).
- `core/apiengine/src/main/kotlin/lava/apiengine/NativeApiEngine.kt` — `ConfigDto/StatusDto + toDto()/toApiStatus()` changed `private` → `internal`, and the defaultJson companion `private` → `internal`, so the JVM regression test exercises the EXACT production serializer path. (Public `ApiEngine/ApiConfig/ApiStatus` API unchanged.)

Fix: apply the project's `kotlinx-serialization` convention plugin so the `@Serializable` DTOs get generated serializers.

Verification test/challenge:

- `core/apiengine/src/test/kotlin/lava/apiengine/ConfigSerializationTest.kt` (cheap JVM, §6.T.1 reproduction). RED with plugin removed: `kotlinx.serialization.SerializationException: Serializer for class 'ConfigDto' is not found ... (4/4 FAIL)`; GREEN with plugin applied:
:core:apiengine:testDebugUnitTest BUILD SUCCESSFUL (13/13).
- Real-device re-run (Pixel_8 / API35 / arm64-v8a, host-direct+HVF cold boot): serialization fix PROVEN — `grep -c SerializationException` over the post-Start logcat is 0 (was the Phase E blocker). C01 PASS. Evidence: `.lava-ci-evidence/phase-e-api-app/2026-06-02-challenge-green-after-serialization-fix.md`.

Remaining defect (separate, out of this task's scope — reported, not fixed): The Challenge re-run surfaced a DISTINCT pre-existing native-linking defect in `lava-api-go/`: the prebuilt Go c-shared `liblavaapi.so` has no `SONAME`, so the JNI bridge `liblavaapi_jni.so` records the absolute host build path (`/Users/.../lava-api-go/build/jniLibs/arm64-v8a/liblavaapi.so`) as its `DT_NEEDED`, which the on-device linker cannot resolve → `dlopen failed` → `embed-start unreachable` → C02/C03/C04 still time out. Remediation (owed to `lava-api-go/`, NOT applied here): add `-ldflags="-extldflags=-Wl,-soname,liblavaapi.so"` to the `go build -buildmode=c-shared` in `lava-api-go/scripts/build-cshared.sh`. Full diagnosis in the evidence file above.

Fix commit: (*this commit*)

Forensic anchor: Phase E real-device Challenge run caught the serialization defect (`.lava-ci-evidence/phase-e-api-app/2026-06-02-phase-e-challenge-execution.md`) — the §6.J/§6.Z bluff class: JVM unit tests green, real `embed-start` broken for every user.

2026-06-02 — cshared-liblavaapi-missing-soname

Root cause: the prebuilt Go c-shared liblavaapi.so (built by lava-api-go/scripts/build-cshared.sh via go build -buildmode=c-shared) had NO DT_SONAME. The JNI bridge liblavaapi_jni.so (lava-api-go/cmd/lavaapi-cshared/jni/CMakeLists.txt) imports it as a CMake IMPORTED_SHARED library via its absolute IMPORTED_LOCATION, so the NDK linker recorded the **absolute host build path** (/Users/.../lava-api-go/build/jniLibs/arm64-v8a/liblavaapi.so) as the bridge's DT_NEEDED. On-device, the Android dynamic linker cannot resolve a host filesystem path → dlopen failed: library "/Users/.../liblavaapi.so" not found: needed by .../liblavaapi_jni.so → LavaNative.nativeStart is unreachable → ApiEngineController.start() maps to Error → the :api-app Challenges C02/C03/C04 time out waiting for "Running". Proven with llvm-readelf -d, NOT guessed. (This is the "remaining defect" the 2026-06-02 serialization-fix entry above recorded as out-of-scope; it is now fixed.)

Affected files:

- lava-api-go/scripts/build-cshared.sh — added the NDK linker soname flag to the c-shared go build: -ldflags="-extldflags=-Wl,-soname,liblavaapi.so".

Fix: give the c-shared .so a real DT_SONAME (liblavaapi.so, relative) so any consumer linking against it records the SONAME — not the absolute host path — in DT_NEEDED. The Android linker then resolves it from the APK's lib dir at dlopen time.

Verification (ELF-level, definitive):

- llvm-readelf -d of all 3 rebuilt ABIs: DT_SONAME = liblavaapi.so present (was absent before). go build EXIT 0 for arm64-v8a / x86_64 / armeabi-v7a; LavaApiStart present in each dynamic symbol table.
- llvm-readelf -d of liblavaapi_jni.so **inside the rebuilt api-app-debug.apk**: (NEEDED) liblavaapi.so (relative) — was the absolute host path. The APK ships both lib/arm64-v8a/liblavaapi.so (53 MB) + liblavaapi_jni.so (25 KB). :api-app:assembleDebug :api-app:assembleDebugAndroidTest BUILD SUCCESSFUL.
- Full evidence: .lava-ci-evidence/phase-e-api-app/2026-06-02-soname-fix-and-containers-gate-blocker.md.

Runtime gate status (honest, §6.Z/§6.J/§6.AG): the C01-04 runtime GREEN through the Containers submodule is OWED on a Containers cmd/emulator-matrix --gradle-module flag — the CLI is hardwired to :app:connectedDebugAndroidTest (submodules/containers/pkg/emulator/{android,containerized}.go) and cannot run :api-app Challenges; that submodule is out of this task's touch-list. The emulator boot itself WAS Containers-orchestrated (--runner=auto→host-direct+HVF, Pixel_8/API35 cold-booted AVD), proven by the runner's [matrix-diag] output. No green was faked; the wrong-module run was stopped before it could false-pass. scripts/run-api-app-challenge-matrix.sh (added this commit) is module-parameterised so the gate runs the moment Containers gains the flag.

Fix commit: (*this commit*)

Forensic anchor: the soname defect was caught by the Phase E real-device Challenge re-run AFTER the serialization fix (`.lava-ci-evidence/phase-e-api-app/2026-06-02-challenge-green-after-serialization-fix.md`) — a native-ELF bluff vector invisible to every JVM/host test: the libraries link + the APK assembles fine on the host, but the embed cannot `dlopen` on the user's device.

archiveorg search pagination halved totalPages (under-reported page count)

Discovered: 2026-06-06 (Completeness Program Phase 4, Stream B), while adding anti-bluff unit tests for the Internet Archive DTO→domain mapping.

Root cause: `SearchResponseDto.toDomain()` computed `totalPages` with page size 100 (`response.numFound / 100`), while the search request issued by `ArchiveOrgSearch` (and `ArchiveOrgBrowse`) uses `rows=50`. `toBrowseResult()` already used the correct `/50`. Because the divisor was double the real page size, `totalPages` for search results was under-reported by ~half — the search UI could not paginate into the back half of any result set with more than 50 hits (e.g. 1234 hits reported 13 pages instead of 25).

Affected files:

- `core/tracker/archiveorg/src/main/kotlin/lava/tracker/archiveorg/feature/ArchiveOrgDto.kt` — `toDomain()` divisor 100 → 50, matching the `rows=50` request and `toBrowseResult()`.

Fix: use page size 50 in `toDomain()` to derive `totalPages`, with a comment anchoring the value to the `rows=50` request in `ArchiveOrgSearch`.

Verification: `ArchiveOrgDtoTest.toDomain` computes total pages by ceiling division of `numFound` by 50 asserts exact `totalPages` for 0/1/50/51/100/101/1234 hits. Falsifiability: re-introducing the `/100` divisor fails that test with the expected vs actual page-count mismatch; reverting to `/50` passes. The test matrix also covers `toBrowseResult` `totalPages` and the conditional metadata map.

Fix commit: (*this commit*)

formatSize rendered comma-decimal separators on non-US locales

Discovered: 2026-06-06 (Completeness Program Phase 3, Detekt correctness backlog).

Root cause: `core/tracker/rutracker/.../domain/Utils.kt:102` `formatSize` used `String.format("%.1f %sB", ...)` without a `Locale`, so the JVM default locale applied. On any comma-decimal locale (`de/ru/fr/...`) a 1.5 MB torrent rendered as "1,5 MB".

Fix: `String.format(Locale.ROOT, ...)`.

Verification: `FormatSizeLocaleTest` forces `Locale.GERMANY/ru-RU` and asserts dot output. Falsifiability: Locale-less format → 2 of 4 tests FAIL (expected "1.5 MB" but was "1,5 MB"). **Fix commit:** bc0a478d.

DownloadServiceImpl cache data race

Discovered: 2026-06-06 (Completeness Program Phase 3, Android leak/race sweep).

Root cause: `DownloadServiceImpl.cache` was a plain `HashMap` read on the caller's coroutine dispatcher (`downloadTorrentFile`) and written from the `DownloadManager BroadcastReceiver.onReceive` (main thread). Concurrent downloads touched a non-thread-safe map across two threads.

Fix: extracted `DownloadUriCache` backed by `ConcurrentHashMap` (pure-JVM, unit-testable per Fifth Law).

Verification: `DownloadUriCacheConcurrencyTest` (16 threads × 500 keys). Falsifiability: `HashMap` backing → test FAILS; `ConcurrentHashMap` → 3/3 green. **Fix commit:** 76fb0879.

ApiEngineController.start() TOCTOU double-bind

Discovered: 2026-06-06 (Completeness Program Phase 3/4).

Root cause: `ApiEngineController.start()` re-entry guard was a read-then-set on `MutableStateFlow`. The controller is a Hilt singleton driven by the `ViewModel` (`Dispatchers.Default`) and the `Service` (`Main.immediate`, `ACTION_RESTART`); two concurrent `start()` calls could both observe `Stopped` and both bind the embed ("address already in use").

Fix: `atomic_state.compareAndSet` guard loop + a `@VisibleForTesting` guard-checkpoint seam (Fifth Law: refactor for testability).

Verification: `ApiEngineControllerTest.concurrent start` parked in the guard gap binds the embed exactly once parks one caller in the read→set gap. Falsifiability: read-then-set → test FAILS (double-bind); `compareAndSet` → 8/8. **Fix commit:** 30e880c6.

TestSearchHistoryRepository.remove kept the matched row (bluff fake)

Discovered: 2026-06-06 (Completeness Program Phase 4, flagged independently by two subagent streams).

Root cause: the shared `:core:testing fake's remove(id)` did filter `{ it.id == id }` — it KEPT the matched row and dropped the others, the inverse of the real `SearchHistoryRepositoryImpl.remove(Room delete(id))`. A Third-Law bluff fake.

Fix: `filterNot { it.id == id }`; added `TestSearchHistoryRepositoryTest`.

Verification: the test asserts `remove(1)` on `[0,1,2]` leaves `[0,2]`. Falsifiability: the shipped bug fails it (expected `[0,2]` but was `[1]`). `feature/search + feature/menu` stay green. **Fix commit:** (*this session*).

On-device API embed could silently drift from the lava-api-go source (no input-tracking)

Discovered: 2026-06-06 (Completeness Program, STREAM API-SYNC).

Root cause: the `core:apiengine` Gradle `buildCshared` task declared ONLY `inputs.file(cSharedScript)` as its input. The `lava-api-go` SOURCE TREE that actually compiles into `liblavaapi.so` was NOT declared as a task input, so Gradle considered the task up-to-date whenever the per-ABI `.so/.h` outputs existed — even after the `lava-api-go` Go source changed. A developer who edited `lava-api-go/internal/...` and re-assembled the `api-app` got the OLD `.so`: the on-device API embed silently drifted from the current API codebase, while every other gate stayed green (a §11.4.69 / §6.J "tests pass, wrong-version feature" bluff vector). There was no mechanism asserting "the packaged embed equals the current API source".

Affected files:

- `core/apiengine/build.gradle.kts` (missing source inputs on `buildCshared`)
- `lava-api-go/scripts/build-cshared.sh` (no source-hash injection / manifest)
- `lava-api-go/internal/version/version.go`, `internal/mobile/mobile.go`
- `core/apiengine/.../ApiEngine.kt`, `NativeApiEngine.kt`

Fix (drift-prevention mechanism):

1. `scripts/compute-api-source-hash.sh` — single source of truth: a deterministic sha256 over the exact non-test `.go` files under `cmd/lavaapi-cshared + internal plus go.mod/go.sum` (the file set that links into the `.so`).
2. `build-cshared.sh` injects that hash into `version.SourceHash` via `-ldflags -X` (the running `.so` reports it through `mobile.Status()`), and writes the committed manifest `core/apiengine/src/main/resources/api-source.hash` on every successful build.
3. `scripts/check-api-app-sync.sh` — CI gate (wired into `scripts/ci.sh`, both `--changed-only` and `--full`): recompute the live hash, compare to the committed manifest, exit 1 loudly on mismatch.
4. `core/apiengine/build.gradle.kts` `buildCshared` now declares the `lava-api-go` source dirs + `go.mod/go.sum` as `@InputFiles`, so Gradle re-runs `build-cshared.sh` on ANY embed-source change (kills the stale-cache drift).

5. On-device proof: api-app Challenge C05

(Challenge05ApiEmbedSourceHashMatchesTest) starts the real embed and asserts `ApiStatus.sourceHash == BuildConfig.LAVA_API_SOURCE_HASH` (no drift), with an empty-hash hard-fail discrimination.

Verification: `lava-api-go/tests/contract/sourcehash_contract_test.go` asserts hash stability, manifest-matches-live, and falsifiability (a content edit to a tracked embed-linked `.go` changes the hash). 3/3 PASS. The gate `scripts/check-api-app-sync.sh` PASSES on the seeded manifest; falsifiability rehearsal: appending a byte to `internal/version/version.go` → gate exits 1 ("on-device API embed is STALE"); revert → exit 0. **Fix commit:** (*this session*).

2026-06-08 — kinozal-magnet-link-declared-but-null (§6.E bluff)

Root cause: `KinozalDescriptor` declares the `MAGNET_LINK` capability, but `KinozalDownload.getMagnetLink()` returned `null` even though the magnet is parsed from the topic/search HTML — a declared-but-empty capability, i.e. a §6.E capability-honesty bluff. The `getMagnetLink` interface method is non-suspend and cannot itself fetch.

Affected files:

- `core/tracker/kinozal/src/main/kotlin/lava/tracker/kinozal/feature/KinozalDownload.kt`
- `core/tracker/kinozal/src/main/kotlin/lava/tracker/kinozal/feature/KinozalTopic.kt`
- `core/tracker/kinozal/src/main/kotlin/lava/tracker/kinozal/magnet/KinozalMagnetCache.kt` (NEW)
- `core/tracker/kinozal/src/main/kotlin/lava/tracker/kinozal/KinozalClientFactory.kt`
- `core/tracker/kinozal/src/test/kotlin/lava/tracker/kinozal/feature/KinozalDownloadTest.kt`
- `core/tracker/kinozal/src/test/kotlin/lava/tracker/kinozal/feature/KinozalTopicTest.kt`
- `core/tracker/kinozal/src/test/kotlin/lava/tracker/kinozal/KinozalClientFactoryCloneUrlTest.kt`

Fix: wire it honestly — a `@Singleton` `KinozalMagnetCache` populated by the real topic-view path (`KinozalTopic.getTopic()`), read back synchronously by `getMagnetLink(id)`; honest `null` until a topic surfaces it. Same pattern `RuTracker` documents for `GetMagnetLinkUseCase`. Cache threaded through the clone-override path.

Verification test/challenge: `KinozalDownloadTest` — "getMagnetLink returns real magnet after the topic page has been viewed". Falsifiability rehearsal (per commit body): reverting `getMagnetLink` to `= null` (the original defect) FAILED with `AssertionError: magnet should be exposed after topic view` (`KinozalDownloadTest.kt:59`); reverted, `core:tracker:kinozal:test` all green (forced `--rerun-tasks`).

Fix commit: 6fa31ad2

2026-06-08 — nnmclub-magnet-link-declared-but-null (§6.E bluff)

Root cause: `NnmclubDescriptor` declares `MAGNET_LINK`, but `NnmclubDownload.getMagnetLink()` returned `null` though both `NnmclubSearchParser` and `NnmclubTopicParser` parse the genuine magnet from NNM-Club HTML — a §6.E bluff.

Affected files:

- `core/tracker/nnmclub/src/main/kotlin/lava/tracker/nnmclub/feature/NnmclubDownload.kt`
- `core/tracker/nnmclub/src/main/kotlin/lava/tracker/nnmclub/feature/NnmclubTopic.kt`
- `core/tracker/nnmclub/src/main/kotlin/lava/tracker/nnmclub/feature/NnmclubSearch.kt`
- `core/tracker/nnmclub/src/main/kotlin/lava/tracker/nnmclub/http/NnmclubMagnetCache.kt` (NEW)
- `core/tracker/nnmclub/src/main/kotlin/lava/tracker/nnmclub/NnmclubClientFactory.kt`
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/feature/NnmclubMagnetExposureTest.kt` (NEW)
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/feature/NnmclubDownloadTest.kt`
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/feature/NnmclubSearchTest.kt`
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/feature/NnmclubTopicTest.kt`
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/NnmclubClientTest.kt`
- `core/tracker/nnmclub/src/test/kotlin/lava/tracker/nnmclub/NnmclubClientFactoryCloneUrlTest.kt`
- `core/tracker/nnmclub/src/test/resources/fixtures/nnmclub/search/search-with-magnet-2026-06-08.html` (NEW fixture)

Fix: wire it honestly — a `@Singleton` `NnmclubMagnetCache` populated by the real topic-view (`NnmclubTopic.getTopic`) and search-row (`NnmclubSearch.search`) paths, read back by `getMagnetLink(id)`; honest `null` when nothing has surfaced it. Cache threaded through the clone-override path. Adds a real fixture (`search-with-magnet-2026-06-08.html`) containing a magnet row.

Verification test/challenge: `NnmclubMagnetExposureTest` (`lava.tracker.nnmclub.feature`). Falsifiability rehearsal (per commit body): reverting `getMagnetLink` to return `null` (the historical bluff) FAILED 2 of 3 — "exposes the parsed magnet after the topic page surfaces it" + "after a search row surfaces it" — `AssertionError` at `NnmclubMagnetExposureTest.kt:50` and `:82`; reverted, `core:tracker:nnmclub:test` 27 tests / 0 failures (forced `--rerun-tasks`).

Fix commit: 13703bc8

2026-06-08 — rutor-magnet-link-declared-but-null + bluff-codifying test (§6.E)

Root cause: `RuTorDescriptor` declares `MAGNET_LINK` but `RuTorDownload.getMagnetLink()` returned an unconditional `null` though `RuTorTopicParser` + `RuTorSearchParser` parse the genuine magnet from `RuTorHTML` — a §6.E bluff. Worse, a pre-existing test "rutor has no synchronous magnet (Capability Honesty)" CODIFIED the bluff as intentional.

Affected files:

- `core/tracker/rutor/.../feature/RuTorDownload.kt`
- `core/tracker/rutor/.../feature/RuTorTopic.kt`
- `core/tracker/rutor/.../feature/RuTorSearch.kt`
- `core/tracker/rutor/.../magnet/RuTorMagnetCache.kt` (NEW)
- `core/tracker/rutor/.../RuTorClientFactory.kt`
- `core/tracker/rutor/.../feature/RuTorMagnetExposureTest.kt` (NEW)
- `core/tracker/rutor/.../feature/RuTorDownloadTest.kt`
- `core/tracker/rutor/.../feature/RuTorSearchTest.kt`
- `core/tracker/rutor/.../feature/RuTorTopicTest.kt`
- `core/tracker/rutor/.../RuTorClientTest.kt`
- `core/tracker/rutor/.../RuTorClientFactoryCloneUrlTest.kt`
- `.gitmodules` + submodules/
 `{doc_processor, llm_orchestrator, llm_provider, llms_verifier, vision_engine}` (the 5 HelixQA dep submodule gitlinks landed in this same commit; see the HelixQA pin entry below)

UNCONFIRMED: the exact `core/tracker/rutor` source-root prefix is abbreviated above; `git show --stat 6d87019d` reports the file basenames but the path column is truncated in the stat output. The basenames + `feature/ / magnet/` subdirs are confirmed.

Fix: wire it honestly via a `@Singleton` `RuTorMagnetCache` populated by the real topic-view + search-row paths, read back by `getMagnetLink(id)`; honest `null` on miss. Threaded through the clone-override path. The bluff-codifying test was replaced with honest-absence semantics.

Verification test/challenge: `RuTorMagnetExposureTest`. Falsifiability rehearsal (per commit body): reverting `getMagnetLink` to return `null` (the historical bluff) FAILED 2 of 3 — "exposes the parsed magnet after the topic page surfaces it" + "after a search row surfaces it" — `AssertionError` at `RuTorMagnetExposureTest.kt:57` and `:97`; reverted, `core:tracker:rutor:test` 76 tests / 0 failures; `RuTorMagnetExposureTest` 3/0/0 (forced `--rerun-tasks`).

Fix commit: 6d87019d

Forensic anchor: §6.E capability-honesty audit, 2026-06-08 session — this one is the sharpest §6.J case of the four because a passing test had been encoding the bluff as the intended contract.

2026-06-08 — gutenbergdishonest-torrent-download-capability (§6.E)

Root cause: GutenbergDescriptor declared TORRENT_DOWNLOAD but the provider serves HTTP e-books (Gutendex EPUB/text/HTML), not .torrent files — a §6.E capability-honesty bluff. The downloaded artifact is not a bencoded .torrent.

Affected files:

- core/tracker/gutenberg/src/main/kotlin/lava/tracker/gutenberg/GutenbergDescriptor.kt
- core/tracker/gutenberg/src/main/kotlin/lava/tracker/gutenberg/GutenbergClient.kt
- core/tracker/gutenberg/src/test/kotlin/lava/tracker/gutenberg/GutenbergCapabilityHonestyTest.kt (NEW)
- core/tracker/gutenberg/src/test/kotlin/lava/tracker/gutenberg/GutenbergClientTest.kt
- core/tracker/gutenberg/src/test/kotlin/lava/tracker/gutenberg/GutenbergDescriptorTest.kt

Fix: mirror archiveorg's established honest pattern — drop TORRENT_DOWNLOAD from the descriptor;
GutenbergClient.getFeature(DownloadableTracker::class) returns null explicitly (the GutenbergDownload impl stays wired but unexposed via the torrent surface). No enum/api change. Stale tests that encoded the bluff were removed.

Verification test/challenge: GutenbergCapabilityHonestyTest pins the contract. Falsifiability rehearsal (per commit body): re-declaring TORRENT_DOWNLOAD + re-gating getFeature so the EPUB-serving impl is again exposed via the torrent-download surface FAILED — "download capability is honest about the artifact it produces" — AssertionError: TORRENT_DOWNLOAD declared but downloaded artifact is not a bencoded .torrent (first byte was 'e', expected 'd') (GutenbergCapabilityHonestyTest.kt:91); reverted, independently re-verified GutenbergCapabilityHonestyTest 2/0/0.

Fix commit: b197f96d

Forensic anchor: §6.E capability-honesty audit, 2026-06-08 session.

2026-06-08 — helixqa-broken-pin-gomod-conflict-markers (broke lava-api-go build)

Root cause: HelixQA's `go.mod` replace directives require 8 sibling own-org (`digital.vasic.*`) Go modules; 3 were present (challenges/containers/security), 5 were MISSING, so the full `helixqa` binary could not build. The broken pinned HelixQA `go.mod` also carried unresolved merge-conflict markers (at lines 124 / 131 / 138 per the commit body), which broke `lava-api-go`'s Go build.

Affected files:

- `submodules/helixqa` (pin bump 5112906 → dd3cf1d)
- `docs/qa/helixqa-dependency-submodules.md` (NEW doc)
- `docs/CONTINUATION.md` (§0 + §3 sync)
- (paired commit 6d87019d adds the 5 dep submodule gitlinks `submodules/{doc_processor, llm_orchestrator, llm_provider, llms_verifier, vision_engine}` from `vasic-digital/{DocProcessor, LLMOrchestrator, LLMProvider, LLMsVerifier, VisionEngine}`)

UNCONFIRMED: the conflict-marker line numbers 124/131/138 are quoted from the commit body, not independently re-diffed in this audit. The pin SHAs 5112906 → dd3cf1d and the `submodules/helixqa` change are confirmed by `git show --stat dd72669b`.

Fix: bump the `helixqa` pin 5112906 → dd3cf1d (clean `go.mod`); add the 5 missing dep submodules (local paths lowercase snake_case per §11.4.29 matching HelixQA's `./doc_processor` replace targets; URLs use the real CamelCase repo names per the must-not-break-technology carve-out).

Verification: physical proof recorded in the commit body — `go build ./cmd/helixqa` exit 0 → 29.7 MB binary; `helixqa version` → v0.2.0; and `lava-api-go` builds (exit 0).

Fix commit: dd72669b (pin bump + doc) — the 5 dep submodule gitlinks landed in companion commit 6d87019d.

Forensic anchor: operator directive ("add all missing dependency Submodules from `vasic-digital/HelixDevelopment`") + §11.4.27/§11.4.28 (own-org deps reachable from root). OWED (tracked in CONTINUATION): §6.W GitLab mirrors for the 5 new submodules; §11.4.29 upstream CamelCase→snake_case repo rename.

2026-06-08 — nav-compose-2.9.0-test-teardown-lifecycle-race

Root cause (confirmed, stack-trace-quoted in the systematic-debugging investigation): `androidx-navigation-compose 2.9.0`'s single-top `NavBackStackEntry` lifecycle reaches `CREATED` asynchronously; the Espresso/Compose test-runner's immediate `Activity.performDestroy` then tries `INITIALIZED → DESTROYED` in one step → `IllegalStateException "State`

must be at least 'CREATED' to be moved to 'DESTROYED'". 2.9.1 release note: "Fixed an issue that caused NavEntries instantiated using single top to never go beyond CREATED in their Lifecycle.State" ([I043ba], b/421095236) — matches exactly (Lava uses launchSingleTop=true).

User impact: NONE — real users get intervening lifecycle pauses. It is a synthetic test-teardown artifact that had forced Challenge tests C04–C08 + deep C11 to be gutted to shallow versions. Blast radius LOW: single direct consumer (core/navigation); -Xcontext-receivers is orthogonal (a Kotlin compiler feature, not the nav lib).

Affected files:

- gradle/libs.versions.toml (pin 2.9.0 → 2.9.1)

Fix: bump androidx-navigation-compose 2.9.0 → 2.9.1.

Verification: ./gradlew :core:navigation:compileDebugKotlin → BUILD SUCCESSFUL with 2.9.1.

UNCONFIRMED / OWED (per commit body): the race-fix confirmation (restoring the deep C04–C08/C11 Challenges + proving the teardown crash is gone) is device-gated — requires the Genymotion VM / a real device, OWED when a device is booted. Not yet verified on-device in this session.

Fix commit: 7e6e7bcb

Forensic anchor: systematic-debugging investigation of the C04–C08/C11 shallow-test regression, 2026-06-08 session.

§6.E — RuTracker MAGNET_LINK declared but getMagnetLink returned null (2026-06-08)

Class: §6.E capability-honesty bluff (declared-but-empty capability).

Symptom: RuTrackerDescriptor declares TrackerCapability.MAGNET_LINK (RuTrackerDescriptor.kt:36) and getFeature(DownloadableTracker) resolves to a non-null RuTrackerDownload, but the only reachable getMagnetLink impl (GetMagnetLinkUseCase) returned null unconditionally for EVERY id on the real stack — even though RuTracker genuinely parses the magnet (TopicMapper.kt:121). A user tapping "Magnet" on a RuTracker topic could never get the magnet the app had already parsed. Identical to the bluff RuTor closed with RuTorMagnetCache.

Root cause: GetMagnetLinkUseCase.invoke(id) = null stub; no path carried the parsed TorrentItem.magnetUri from the topic/search fetch to the synchronous DownloadableTracker.getMagnetLink. Audit: docs/qa/magnet-label-honesty-audit-2026-06-08.md (W4a).

Affected files:

- core/tracker/rutacker/.../magnet/RuTrackerMagnetCache.kt (new — RuTor pattern)

- `core/tracker/rutacker/.../domain/GetMagnetLinkUseCase.kt` (reads the cache)
- `core/tracker/rutacker/.../feature/RuTrackerTopic.kt` (populates on `getTopic`)
- `core/tracker/rutacker/.../feature/RuTrackerSearch.kt` (populates per result row)
- `core/tracker/rutacker/.../RuTrackerSubgraphBuilder.kt` (shared cache, clone path)
- `core/tracker/client/.../di/TrackerClientModule.kt` (Hilt `@Provides` reads the cache)

Fix: adopt the RuTor magnet-cache pattern — a `@Singleton` process-lifetime `RuTrackerMagnetCache` populated by `RuTrackerTopic.getTopic` / `RuTrackerSearch.search` from the already-mapped `magnetUri`, read by `GetMagnetLinkUseCase`. Honest null preserved for ids never surfaced.

Verification (§6.T.1 reproduction-before-fix):

- RED (pre-fix, against the stub): `RuTrackerMagnetExposureTest > getMagnetLink` exposes the magnet the topic mapper surfaced FAILED — "§6.E BLUFF: ... `getMagnetLink` returned null".
- Mutation rehearsal (post-fix, `invoke → null`): same test FAILED again (exit 1).
- GREEN (fix): `RuTrackerMagnetExposureTest` 2/2 PASS;
:core:tracker:client:testDebugUnitTest BUILD SUCCESSFUL (Hilt graph valid; enumeration gate 5/5 + 8/8).

Forensic anchor: SA4 §6.E honesty audit, 2026-06-08 parallel-fleet session. Full-loop (topic fetch → cache → download) covered on-device by the restored C05/C06 Challenges (device run owed).

§6.Q/C37 — CrossTrackerFallbackModal dead-ended at the screen layer (2026-06-08)

Class: dead-UI (capability present in code, unreachable by users — the §6.Q/C37 class).

Symptom: `feature/search_result` has a full cross-tracker-fallback feature — `CrossTrackerFallbackModal` composable + `SearchResultViewModel.proposeFallback/ onFallbackAccept/ onFallbackDismiss` + `SearchResultAction.FallbackAccept/ FallbackDismiss` + `SearchResultState.crossTrackerFallback` — BUT `SearchResultScreen` never read `state.crossTrackerFallback` and never called `CrossTrackerFallbackModal(...)`. A real user whose search failed on one tracker could never see the "Try " offer. Found by the SA5 Challenge restoration audit (C07/C08 had to be written as contract guards, not the real flow, because the real flow was unreachable).

Affected files:

- `feature/search_result/.../SearchResultScreen.kt` (renders the modal)

- app/src/androidTest/.../challenges/
Challenge07CrossTrackerFallbackAcceptTest.kt
- app/src/androidTest/.../challenges/
Challenge08CrossTrackerFallbackDismissTest.kt

Fix: in SearchResultScreen, render

CrossTrackerFallbackModal(failedTracker, proposedTracker, onAccept = { onAction(FallbackAccept) }, onDismiss = { onAction(FallbackDismiss) }) when state.crossTrackerFallback != null (display names via the existing state.providerDisplayNames). C07/C08 upgraded from contract guards to real rendered-modal Compose tests (C30 pattern): render → assert "RuTracker is unavailable" + "Try RuTor"/"Cancel" → tap → assert the callback fires.

Verification: :feature:search_result:compileDebugKotlin + :app:compileDebugAndroidTestKotlin BUILD SUCCESSFUL. §6.J falsifiability (KDoc): delete the state.crossTrackerFallback?.let{} block → dead-end returns; or change the confirm/dismiss label → the rendered test's onNodeWithText finds no node → assert fails. Device-run of C07/C08 owed on the VM (network-independent).

Forensic anchor: SA5 Challenge-restoration finding, 2026-06-08 parallel-fleet session.

§6.J DEVICE-GATE CORRECTION — nav-compose 2.9.1 does NOT fully fix the NavBackStackEntry teardown crash (2026-06-08)

Correcting the 2.9.0→2.9.1 entry (commit 7e6e7bcb). That bump claimed to fix the test-teardown NavBackStackEntry lifecycle race and explicitly marked the device confirmation OWED. Device confirmation is now done — on the Genymotion Pixel 9 VM (API 35) — and it **FALSIFIES the claim** for the search_input route: Challenge11Archive0rgAnonymousSearchTest crashes the app at MainActivity destroy with IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED' on the search/search_input?... NavBackStackEntry (LifecycleRegistry.kt:92). 2.9.1 was necessary but NOT sufficient.

Status: OPEN. C00/C01/C07/C08 PASS on the VM (6/7); C11 RED. NOT bluffed green. Incident: .lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json. Evidence: .lava-ci-evidence/genymotion/9e7505b3-fleet-run/.

Next: dedicated systematic-debugging cycle — evaluate a newer androidx-navigation OR a NavHost/teardown lifecycle guard, then device re-verify.

api-source.hash manifest staleness — §6.A contract test RED→GREEN (2026-06-08)

Commit 7ce9c2b2 edited two embed-linked lava-api-go sources (internal/cache/cache.go + internal/storage/sqlite.go) without regenerating the API[?]embed sync manifest, so TestSourceHash_ManifestMatchesLive (lava-api-go/tests/contract) failed: committed 95c36d1773362cee2b90db398f2e2827586fef83071ba3feb12ce51e02e12b9d != live 70ebb53a1a9c3365f651ae7ae651dd4404d58976b97b3b2589db39f4d7752470. Root cause: the manifest at core/apiengine/src/main/resources/api-source.hash was last written at e153d7c0 (2026-06-06), pre-dating 7ce9c2b2. The contract test correctly caught real source/embed drift — exactly its §6.A purpose.

Fix: regenerated api-source.hash to the live scripts/compute-api-source-hash.sh value (the canonical mechanism build-cshared.sh uses). NOT a test edit — the regeneration the test's own error message prescribes. Verified: go test ./tests/contract -run TestSourceHash → 3/3 PASS (incl. the falsifiability sibling ...ContentEditChangesHash); full go test ./... → exit 0 (39 packages ok, real Postgres-in-podman e2e ran).

YTS curated provider unreachable — yts.mx domain rotated out of DNS (2026-06-13)

Symptom: the YTS curated provider (shipped 2026-06-12, commit 462a0308) passed all fixture unit tests but its LIVE call failed: live Search: yts: provider: unknown error. Surfaced by a main-stream §6.J re-verification (running the -tags realtrackers live tests instead of trusting the prior commit-body claim).

Root cause: the provider hardcoded a single base URL https://yts.mx. As of 2026-06-13 yts.mx has NO A record on public DNS (confirmed dig @1.1.1.1 and @8.8.8.8 both return <none>) — YTS rotated its domain since the provider shipped. The fixture parser was correct (unit tests green) while the feature was broken for real users (§6.G/§6.L fixture-green / feature-broken). The current live canonical host is yts.bz (yts.lt / yts.am 301-redirect to it).

Affected files: lava-api-go/internal/provider/curated/yts/{client.go,provider.go, live_realtrackers_test.go,yts_test.go}.

Fix: mirror failover — Client now holds a list of base URLs (DefaultBaseURLs = yts.mx, yts.bz, yts.lt, yts.am) tried in order; the first successful answer wins; each attempt is bounded by perAttemptTimeout (8s) so a dead/hanging mirror cannot stall the search. NewClient(single) preserved as the httpstest seam; production New() uses NewClientWithMirrors(DefaultBaseURLs). The live test now queries a real movie TITLE (interstellar; 1080p matches no movie and legitimately returns 0) across the mirror list.

Verification: new TestSearch_FailsOverToHealthyMirror (falsifiable: break the failover loop → Search across [dead, healthy] mirrors should fail over, got error: yts: HTTP 503;reverted) +

TestSearch_AllMirrorsDownSurfacesError. Live -tags realtrackers PASS in 0.559s (fast failover past dead yts.mx to a live mirror, real magnets). Full go test ./... → 0 FAIL; embed source-hash regenerated 0d7da603; check-api-app-sync.sh GREEN. Evidence: .lava-ci-evidence/bluff-hunt/2026-06-13-yts-domain-failover.json.

Follow-up: TPB (apibay.org) + Torrents-CSV (torrents-csv.com) still hardcode a single domain — same rotation-risk class; mirror-failover hardening noted as a follow-up.

Crashlytics remediation sweep — 6 open issues (2026-06-13)

Operator directive: investigate ALL open Crashlytics issues, systematic-debug, fix/improve, cover with falsifiable validation tests + §6.O closure logs. Real data pulled from Firebase (project lava-vasic-digital, release app). Per-issue:

- **7df61fdb NON_FATAL JobCancellationException (8 events, 1.2.21) — FIXED.** Catch-site rethrow (defense-in-depth over the prior sink-level filter). New Throwable.rethrowIfCancellation() in core/common/.../analytics/CancellationRethrow.kt called first in every coroutine catch/onFailure that records a non-fatal across 8 ViewModels (onboarding, login, menu, search_input, search_result, topic, bookmarks, favorites). Root cause: broad catch (e: Exception) swallowed Kotlin's structured-concurrency CancellationException — both telemetry pollution AND broken cooperative cancellation. Test: core/common/.../CancellationRethrowTest.kt (5 tests). Bluff-Audit: no-op the rethrow → 3 tests RED at CancellationRethrowTest.kt:75; reverted. Closure: .lava-ci-evidence/crashlytics-resolved/2026-06-13-jobcancellation-catch-site-rethrow.md.
- **40a62f97 FATAL painterResource layer-list (1.2.19) — VERIFIED already covered.** LavaIconsAppIconColorRegressionTest + closure 2026-05-14-welcome-layerlist-painter-crash.md already in place. No change.
- **39469d3b FATAL okhttp schemeless URL "djdnd" (1.2.21) — FIXED (owed regression test added).** The prior closure admitted the use-case test was "owed in a follow-up"; this is it. New ProbeMirrorUseCaseTest.schemeless URL returns Unreachable instead of crashing (real ProbeMirrorUseCase + real OkHttpClient). Bluff-Audit: change the catch (IllegalArgumentException) to a non-matching type → test RED with the leaked IllegalArgumentException at :80; reverted. Closure: .lava-ci-evidence/crashlytics-resolved/2026-06-13-probemirror-schemeless-url-regression.md.
- **042b9b61 NON_FATAL CertPath trust-anchor (1.3.3) — VERIFIED already covered.** Fixed at 1.3.4 (commit 0deb54e7); ProviderCatalogRepositoryTest uses a self-signed-TLS MockWebServer; closure 2026-06-12-provider-catalog-fetch-tls.md present. No change.
- **6519b490 NON_FATAL rutracker parseUserId "user-id not found" (1.2.22) — WORKING-AS-DESIGNED (no production change).** New GetCurrentProfileUseCaseUserIdTest proves the 4 production selectors

DO parse a logged-in page (not stale) and the non-fatal fires only on a genuine guest/expired page. The logged-in test is the future staleness canary. Bluff-Audit: reduce selectors to a non-matching one → canary RED at :100; reverted. Closure: `.lava-ci-evidence/crashlytics-resolved/2026-06-13-rustracker-parseuserid-guest-page.md`.

- **3937b7f0 NON_FATAL SSE "Unable to resolve host lava-api.local" (1.3.0)** — **IMPROVED.** `SearchResultViewModel.applySseError` now classifies connectivity-class reasons (host-resolve / refused / timeout) as a lower-severity `recordWarning("sse_endpoint_unreachable")` instead of a crash-feed non-fatal; genuine backend errors stay non-fatals. User-visible Error + Retry UX unchanged. New `String.isConnectivityFailure()` helper. Test: `SearchResultSseConnectivityTelemetryTest` (2 tests). Bluff-Audit: force the classifier to `if (false)` → host-resolve test RED; reverted. Closure: `.lava-ci-evidence/crashlytics-resolved/2026-06-13-sse-host-resolve-telemetry-severity.md`.

Affected production files: `core/common/.../analytics/CancellationRethrow.kt` (new); 8 feature ViewModels (rethrow calls); `core/common/build.gradle.kts` (coroutines-test dep); `feature/search_result/.../SearchResultViewModel.kt` (SSE severity classifier). `ProbeMirrorUseCase.kt` + `rustracker GetCurrentProfileUseCase.kt` UNCHANGED (already-correct fixes; tests added).

§6.AC/§6.O — JobCancellation non-fatal noise: download catch-site rethrow (2026-06-13)

Crashlytics: `7df61fdb64f9928b067624d6db395ca` (NON_FATAL, `kotlinx.coroutines.JobCancellationException` — "StandaloneCoroutine was cancelled", 8 events / 1 user / first=last 1.2.21; not recurring on 1.3.x).

Root cause. `DownloadServiceImpl.downloadHttpFile` (`core/downloads`) — a `suspend fun` — wrapped its write path in a broad catch (`t: Throwable`) that called `analytics.recordNonFatal(t, ...)`. When the download is abandoned mid-write (calling scope cancelled — user left the screen / ViewModel cleared), `CancellationException` was swallowed AND recorded as a non-fatal: false telemetry (issue `7df61fdb` dashboard noise) plus broken cooperative cancellation (the catch returned `null` instead of letting the cancellation propagate). A grep of every telemetry-recording catch across `core/` `feature/` `app/` `main` sources confirmed this was the ONLY remaining offending site — all 8 feature ViewModels + `LavaApplication` already call `rethrowIfCancellation()`.

Fix. `t.rethrowIfCancellation()` (the existing shared `lava.common.analytics` helper — reused, not re-created) is now the FIRST statement of the `downloadHttpFile` catch, before `recordNonFatal`.

Verification test. `core/downloads/src/test/.../DownloadServiceCancellationTest.kt` (new) — cancellation during http write is rethrown and never recorded as non-fatal. Real `DownloadServiceImpl` + recording `AnalyticsTracker`; only the outermost Android boundary (`Context` / the `Environment` static the write path uses) is faked to throw `CancellationException` mid-write. Asserts the cancellation PROPAGATES and the recorded-non-fatal count is 0. Bluff-Audit:

delete the rethrow line → `AssertionError: cancellation must propagate, not be swallowed into a null return`
(`DownloadServiceCancellationTest.kt:87`); reverted → BUILD SUCCESSFUL.

Affected files: `core/downloads/.../DownloadServiceImpl.kt` (rethrow), `core/downloads/build.gradle.kts` (mockk + coroutines-test test deps + `returnDefaultValues`), `core/downloads/.../DownloadServiceCancellationTest.kt` (new). Closure log: `.lava-ci-evidence/crashlytics-resolved/2026-06-13-cancellation-noise-7df61fdb.md`. Queued for the 1.3.7 release cycle (version files NOT touched).

chosen online server appears TWICE in Settings → Server list (2026-06-14)

Symptom (operator-reported): "When we open settings Server list, the chosen online server appears TWICE in the list." The user onboarded with an online/cloud API endpoint; in Settings → the server/connection list, that chosen endpoint is shown duplicated.

Root cause: `core/data/src/main/kotlin/lava/data/converters/Endpoint.kt:69` builds the Room PRIMARY KEY id for an `Endpoint.GoApi` as `GoApi(${packHost()})`, and `packHost()` (`Endpoint.kt:52-60`) appends the additive `key/platform/storage` fields after a `#` sentinel. The SAME physical server (same `host:port`) reaches the persisted list via TWO paths that differ only in those additive fields:

- the cloud "Add server" flow (`feature/onboarding/.../OnboardingViewModel.kt:179` `onAddCloudApi` → `CloudApiDefaults.parse`) builds a BARE `GoApi(host, port)` (`key=null, platform=null, storage=null`) → `id GoApi(host:port)`;
- the on-device / mDNS-discovered flow (`OnboardingViewModel.kt:710` `onOnDeviceApiReturned` carries a per-instance key; `OnboardingViewModel.kt:239` `startApiDiscovery` carries `platform+storage` TXT attributes) builds a KEYED `GoApi` → `id GoApi(host:port#k=...)`.

Both call `endpointsRepository.add()` → `EndpointDao.insert(OnConflictStrategy.REPLACE)`. `REPLACE` de-dups ONLY on a matching primary-key id, so the two different ids produce TWO Room rows for the one server. `EndpointsRepositoryImpl.observeAll()` mapped the DAO list straight through with no de-dup by server identity, so the `Connections/Server list` (`feature/connection/.../ConnectionsViewModel.kt:79` `observeConnections` → `ObserveEndpointsStatusUseCase`) rendered the same `host:port` twice.

Affected files: `core/data/src/main/kotlin/lava/data/impl/repository/EndpointsRepositoryImpl.kt` (fix), `core/data/src/test/.../EndpointsRepositoryImplFilterTest.kt` (tests).

Fix: `EndpointsRepositoryImpl.observeAll()` now applies `.distinctBy(::serverIdentity)` after the model-map / RuTracker-filter step. `serverIdentity(GoApi)` is `"GoApi(host:port)"` — the transport-defining identity, deliberately EXCLUDING the additive key/platform/storage fields that bloat the Room id — so one row shows per actual server regardless of which path added it. `distinctBy` keeps the first occurrence (Room emits in insertion order). Per-endpoint auth is unaffected: the active endpoint's own key is persisted separately by `lava.securestorage.model.EndpointConverter`, not by this list path.

Verification test/challenge:

`EndpointsRepositoryImplFilterTest.observeAll_deduplicates_same_server_added_via_two_paths (new)` — inserts the same host:port via a bare `GoApi` and a keyed `GoApi`, asserts the emitted list contains that server exactly once. Companion `observeAll_keeps_distinct_servers` guards that two genuinely different servers both survive (the de-dup is by identity, not a blanket collapse). Bluff-Audit: with the `.distinctBy(::serverIdentity)` line removed, the test FAILED `java.lang.AssertionError: The chosen online server MUST appear exactly ONCE in the Server list, not twice (operator defect 2026-06-14) expected:<1> but was:<2>`
(`EndpointsRepositoryImplFilterTest.kt:125`); restored → BUILD SUCCESSFUL (5/5).

Fix commit: this commit. **Forensic anchor:** operator-reported 2026-06-14; reproduced before fix per §6.T.1. Version files NOT touched.

search "Something went wrong" across providers via the on-device/LAN API (2026-06-14)

Symptom (operator-reported): onboarded RuTracker + YTS + Kinozal, searched "prince" on Home with all 3 selected, using the Android (on-device) API as the endpoint → "Error: Something went wrong, please try again." Every provider failed.

Root cause (CONFIRMED — same class as SP-3.1 / Defect-A, fixed for / providers but never for per-provider ops): `ApiBackedTrackerClient` per-provider requests (`search/getString/getBytes/postJson/healthCheck`) were built on the UNQUALIFIED, strict, system-trust `OkHttpClient` and carried NO per-instance auth key. The on-device / LAN API is self-signed (so the system-trust client fails the TLS handshake) AND auth-gated by the `Lava-Auth` header (so even on plain HTTP it 401s without the per-instance key). Either way the SDK surfaced the generic "Something went wrong." `NetworkModule.kt:101` documents that the unqualified client "MUST NOT be used for LAN endpoints"; only the `@Named("lan")` permissive client may. /providers had been migrated to the LAN client + key, but per-provider ops had not.

Affected files: `core/tracker/client/.../ApiBackedTrackerClient.kt` (+ `withAuth()` on every request), `.../ApiBaseUrlHolder.kt` (per-endpoint key), `.../di/TrackerClientModule.kt` (factory now injects `@Named("lan")` client + `authFieldName + ApiBaseUrlHolder.currentKey()`), `feature/onboarding/.../OnboardingViewModel.kt` (`ApiBaseUrlHolder.set(url,key)`), `core/domain/.../RepopulateProvidersOnStartupUseCase.kt` + `app/.../StartupProvidersModule.kt` (cold-start key thread-through).

Fix: the active-API base URL is now stored WITH its per-instance key in `ApiBaseUrlHolder`; `TrackerClientModule` builds every `ApiBackedTrackerClient` with the `@Named("lan")` permissive-TLS client and threads `authFieldName+authKey`; `withAuth()` attaches `Lava-Auth: <key>` to every per-provider request. So search/browse/download against a self-signed, authenticated on-device API succeed instead of failing the handshake or 401ing.

Verification test/challenge:

- Unit (reproduction):
`ApiBackedTrackerClientTest.search_attachesPerEndpointAuthKey_so AuthGatedApiReturnsResults` (on-device `MockWebServer` 401s without `Lava-Auth: k`, returns results with it)
 - `search_withoutAuthKey_throwsOnAuthGatedApi`. Bluff-Audit: dropping `.withAuth()` → search 401s → test FAILED; restored → GREEN.
- Wiring contract: `LanHttpClientWiringContractTest` reflectively asserts the factory's `OkHttpClient` param is `@Named("lan")` (closes the gap that nothing asserted the registry was wired with the permissive client).
- Device: `Challenge44ApiSearchAuthTest` (on-device `MockWebServer`) — PASS on the Genymotion VM in the 1.3.8-1065 client gate (BUILD SUCCESSFUL).

Fix commit: landed in the 1.3.8-1065 cycle (search-auth thread-through). **Forensic anchor:** operator-reported 2026-06-14; reproduced before fix (§6.T.1).

onboarding "Pick your providers": select-all / deselect-all control (2026-06-14)

Symptom (operator request): "On pick your providers onboarding screen we must have check and uncheck all checker, especially if there are multiple choices in dozens." No way to toggle every provider at once.

Fix: added `OnboardingAction.ToggleAllProviders + OnboardingViewModel.onToggleAllProviders()` (computes a single target = ! (all currently selected) and maps every provider to it — so it acts as select-all when not all are selected, deselect-all when all are). `ProvidersStep` renders a `select_all_providers`-tagged control when `providers.size >= 2`. **Affected files:** `feature/onboarding/.../OnboardingAction.kt`, `OnboardingViewModel.kt`, `steps/ProvidersStep.kt`, `OnboardingScreen.kt`. **Verification:** `Challenge41OnboardingSelectAllProvidersTest` — PASS on the Genymotion VM (1.3.8-1065 gate). Drives the real `ProvidersStep`, taps select-all, asserts all selected; taps again, asserts all cleared. **Fix commit:** 1.3.8-1065 cycle.

onboarding provider Configure: password field masking + eye toggle (2026-06-14)

Symptom (operator request): "password field MUST behave like every regular password field — masking letters, with eye icon to show password." The Configure step's password was plain text.

Fix: ConfigureStep password field now uses `visualTransformation = if (passwordVisible) None else PasswordVisualTransformation()`

- `KeyboardType.Password` + a trailing eye control (`password_visibility_toggle`, content descriptions "Show password"/"Hide password") that flips `passwordVisible`. **Affected files:** `feature/onboarding/.../steps/ConfigureStep.kt`. **Verification:** `Challenge42OnboardingPasswordMaskingTest` — PASS on the Genymotion VM (1.3.8-1065 gate). NOTE (anti-bluff): Compose retains the RAW text in `EditableText` regardless of the visual mask, so `assertDoesNotExist(plaintext)` is INVALID — the test asserts the masking STATE via the eye control's content-description round-trip (Show→Hide→Show). The original `assertDoesNotExist` form FAILED on device; the device run caught the test bug (a real \$6.Z win) and it was rewritten to the content-description contract. **Fix commit:** 1.3.8-1065 cycle (test corrected in c366454f).

api-app foreground service crashes after ~6h — dataSync → specialUse (2026-06-14)

Symptom (Crashlytics FATAL):

`ForegroundServiceStartNotAllowedException` + `ForegroundServiceDidNotStopInTimeException` on the long-lived on-device API server after ~6h uptime (Galaxy S23 Ultra / Android 16). Issues `9ba8502ee0ba0d1fdd03987650b8acf8` + `b9baeaede585fc3bc9b515c27cde532c`.

Root cause (CONFIRMED): `ApiEngineService` used `foregroundServiceType="dataSync"`, which Android 14+ caps at ~6h cumulative runtime/24h; the long-lived LAN API server exhausts the budget and `startForeground` then throws.

Fix (operator-approved specialUse): `foregroundServiceType dataSync` → `specialUse` (no time budget) + `FOREGROUND_SERVICE_SPECIAL_USE` permission + `PROPERTY_SPECIAL_USE_FGS_SUBTYPE` (Play-review justification). Defensive: `startForeground` wrapped in try/catch (`ForegroundServiceStartNotAllowedException`) → graceful `stopSelf`, and an `onTimeout(startId, fgsType)` override stops cleanly — so even a future budget degrades gracefully instead of crashing. **Affected files:** `api-app/src/main/AndroidManifest.xml`, `api-app/.../service/ApiEngineService.kt`. Design: `docs/issues/2026-06-14-apiapp-fgs-datasync-budget-fix-design.md`. **Verification:** `:api-app:compileDebugKotlin` SUCCESSFUL + the 0.2.8-12 release FGS `specialUse` cold-start canary on the Genymotion VM (the service starts under `specialUse` with no FGS exception — a misconfigured type would throw at `startForeground` → FATAL). \$6.O closure log: `.lava-ci-evidence/crashlytics-resolved/2026-06-14-apiapp-fgs-datasync-budget.md`. **Fix commit:** `ed03cac2` (cherry-picked to master from 10f39f43).

Defect B — api-app Crashlytics telemetry misattributed to the CLIENT Firebase app (2026-06-14)

Symptom: api-app crashes/non-fatals (incl. the FGS FATALs above) landed under the CLIENT Firebase app's dashboard, not the api-app's own app — so api-app telemetry was indistinguishable from client telemetry.

Root cause (CONFIRMED via Firebase MCP): the tracked (gitignored) api-app/google-services.json mapped the api-app package names (digital.vasic.lava.api / .api.dev) to the CLIENT app's mobilesdk_app_id values. The google-services Gradle plugin matches the client[] entry by package_name, so the api-app build baked in the client app id and registered as the client app.

Fix: replaced api-app/google-services.json (gitignored, on-disk only — never committed per §6.H) with the correct config mapping the api-app packages to the real "Lava API (release)" ...d57b960e955645f6cfd20a / "Lava API (debug)" ...2932451e07ca80a7cfd20a app ids, fetched via Firebase MCP from project lava-vasic-digital (both api-app Firebase apps already exist — no creation needed). api-app rebuilt + re-staged for 0.2.8-12. **Affected files:** api-app/google-services.json (gitignored secret — NOT committed). Findings: docs/issues/2026-06-14-defect-b-telemetry-attribution-findings.md. **Verification:** api-app rebuilt SUCCESSFUL with the corrected config + the FGS cold-start canary passes on the rebuilt artifact. Once 0.2.8-12 is distributed + observed, api-app crashes will appear under the api-app dashboard (operator confirms post-distribute). **Fix commit:** this cycle (config replaced on disk; findings doc fb0bfec9).

search does not work in any scenario — 5-layer cascade (2026-06-14)

STATUS (honest, §6.T.1/§11.4.6): search does NOT yet work on-device. Layers 1+2+3 are fixed (L1+L2 device-verified, L3 RED→GREEN at the test layer). Layer 4 is proven a TEST-VM artifact (production grant is safe — clean matched-pair run showed granted=true). **Layer 5 is OPEN/VERIFYING** — even on a clean matched pair with the permission granted and the engine running, /v1/{provider}/search STILL 401s; the Lava-Auth key is not reaching the search request. A logcat-instrumentation run is in progress. **1.3.9 (client-only) is NOT shipped** until the L5 on-device gate is GREEN. api-app unchanged at 0.2.8-12. This entry was extended from 3 to 5 layers as the cascade was peeled back; do not read any "fixed" below as "search works" — it does not yet.

Symptom (operator-reported, 1.3.8-1065): "search still does not work in any scenario." Searching any query against the chosen on-device / LAN API returned no results across every provider. The HelixQA video-QA agent could not produce a search-results walkthrough because search genuinely failed.

Root cause (CONFIRMED — a CASCADE of three independent bugs, all proven from on-device evidence, not one):

- **Layer 1 — onboarding never persisted the chosen API to settings.endpoint.** OnboardingViewModel.onSelectApi wrote the selected API to the Room Endpoint list + ApiBaseUrlHolder (what GET /providers + the dynamic SDK clients read) but NEVER to settings.endpoint. The home search resolves its target host from SettingsRepository.getSettings().endpoint (NetworkApiRepositoryImpl.endpoint()), which therefore stayed at the default Endpoint.GoApi("lava-api.local") → UnknownHostException (the host never resolves on the LAN). So search targeted the wrong host even though /providers worked.
- **Layer 2 — GoApi search routed to GET /v1/search, a route NO backend serves.** SearchResultViewModel's dispatch sent every GoApi multi-provider search to observeSseSearch → GET {base}/v1/search. Verified across the whole Go codebase: internal/router/router.go serves /providers + /v1/:provider/{op} + /jackett/search + /health/ready; the standalone serves legacy /search + /v1/{provider}/search; **neither registers /v1/search** → 404 for every GoApi user.
- **Layer 3 — mDNS-discovered endpoint persisted without its Lava-Auth key (FIXED, 21031f2a).** onSelectApi adopted a discovered API's host/port but never attached the locally-readable key, so the persisted settings.endpoint was keyless → /v1 ops 401 even when the key was readable. Fix: onSelectApi reads the local api-app key for keyless mDNS endpoints + an app-side cold-start key-restore heals an already-persisted keyless endpoint. Both RED→GREEN at the test layer.
- **Layer 4 — READ_API_KEY signature permission not granted on the test VM (TEST-VM ARTIFACT, production-safe).** After L3, the key reader still returned null because the signature permission digital.vasic.lava.permission.READ_API_KEY was not granted to the client on the VM. Root cause (proven, full evidence in docs/issues/2026-06-14-readapikey-permission-variant-analysis.md): the permission name is a **fixed literal** (no .dev suffix) while every other cross-app identifier is variant-suffixed, and the VM had BOTH differently-signed api-app variants (debug ...api.dev + release ...api) in its install history → the Android INSTALL_FAILED_DUPLICATE_PERMISSION trap denies the grant. **Production is SAFE** — the shipped release-client + release-api-app pair shares one signing cert and grants the permission at install; **a clean matched-pair install confirmed granted=true on-device.** Real users never co-install debug
 - release variants. A MEDIUM-priority QA-fidelity hardening (variant-suffix the permission name, like the authority already is) is recommended so the test VM exercises the production grant path — not shipping-blocking.
- **Layer 5 — Lava-Auth key not reaching the ApiBackedTrackerClient search request (OPEN / VERIFYING — the load-bearing open layer).** On a CLEAN matched pair (release client + release api-app, same signature, granted=true, engine running, fresh onboard), /v1/{provider}/search STILL returns 401. The key is granted and readable but is not reaching the search request. **Evidence caveat:** a public GET /providers → 200 proves reachability ONLY — /providers is registered BEFORE the auth middleware, so a 200 there does NOT prove the key is present/valid; only an authenticated /v1/{provider}/... exercises the key. A logcat-instrumentation run is in progress to pinpoint exactly which link in the key-flow drops the key. **Until this closes,**

search does NOT work on-device and no layer above it makes search functional on its own.

Affected files:

- `feature/onboarding/src/main/kotlin/lava/onboarding/OnboardingViewModel.kt` (`onSelectApi` now calls `SetEndpointUseCase` to persist the active `settings.endpoint`; nullable injected seam, real impl bound by Hilt).
- `core/domain/src/main/kotlin/lava/domain/usecase/RepopulateProvidersOnStartupUseCase.kt` (`reconcileActiveEndpoint()` heals existing installs whose onboarding predated the Layer-1 fix — adopts the Room GoApi when `settings.endpoint` is an orphan default).
- `feature/search_result/src/main/kotlin/lava/search/result/SearchResultViewModel.kt` (GoApi multi-provider search re-routed from `observeSseSearch` → `/v1/search` to `observeStreamMultiSearch` → `sdk.streamMultiSearch` → GET `/v1/{provider}/search`, the served path that carries the per-instance Lava-Auth key + permissive-LAN client from the 1.3.8 wiring). The dead `observeSseSearch` → `/v1/search` consumer + its 3 SSE bluff tests were deleted in `addaacd0`.
- `feature/onboarding/.../OnboardingViewModel.kt` + `MainActivity` key-restore (Layer 3, 21031f2a): `onSelectApi` reads the local api-app key for keyless mDNS-discovered endpoints; cold-start key-restore heals a persisted keyless endpoint.

Verification tests added (real-stack reproduction, falsifiability-rehearsed):

- `OnboardingViewModelDynamicProvidersTest` — selecting an API persists it as the active settings endpoint the search path reads (real `ViewModel` + `SetEndpointUseCase` seam; reproduces Layer 1: without the persist write, the active endpoint stays the orphan default).
- `RepopulateProvidersOnStartupUseCaseTest` — cold start heals a stale orphan active endpoint to the onboarded server in the list (real use case; reproduces the existing-install heal path).

On-device Chucker evidence (1.3.8 debug client, Genymotion VM): GET `https://10.0.3.16/providers` → 200; GET `https://lava-api.local/search?query=prince...` → `UnknownHostException` (host never resolves; api-app "Requests served: 0"). Traced via the flow-db Endpoint row + `chucker.db` transactions; surfaced by the HelixQA video-QA agent.

§6.J bluff removed: `SearchResultSseErrorRetryTest` drove `observeSseSearch` over a `MockWebServer` that SERVED `/v1/search` — the route no backend registers — and asserted SSE error/retry rendering. It passed green while the real feature 404'd (Seventh Law clause 4: mock served the endpoint production 404s). Removed. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-06-14-sse-search-v1search-unserved-bluff.json`.

Fix commits: `d05bc71e` (L1+L2), `21031f2a` (L3 key restore), `addaacd0` (L4 cleanup — dropped ignored authority param + dead SSE path).

§6.J meta-lesson (this cycle): a verification subagent proposed an L4 fix — a hardcoded `".keyprovider"` authority string — and the main stream REJECTED it after reading the source: `MainActivity.buildApiKeyReader` returns { →

`client.read()?.key }`, which takes no authority arg and ignores any passed. The proposed change would not have touched the real grant condition (the signature match) → a §6.J bluff (and a §6.R hardcoded literal). Verify subagents root-cause; the main stream verifies a proposed change reaches the failing path before applying.

REMAINING / OPEN / VERIFYING (honest, per §6.T.1 / §11.4.6 — search does NOT yet work on-device):

- **Layer 5 (OPEN, top priority):** clean matched pair, granted=true, engine up, fresh onboard → `/v1/{provider}/search` STILL 401; the Lava-Auth key is read but not delivered on the search request. Logcat-instrumentation run in progress to find the drop point. **1.3.9 (client-only) NOT shipped until this gate is GREEN.** api-app unchanged at 0.2.8-12.
- **Layer 4 (OWED, MEDIUM/QA-fidelity):** variant-suffix the `READ_API_KEY` permission name so the test VM exercises the production grant path; production grant is already safe. Not shipping-blocking.
- Existing installs that do NOT re-onboard get a keyless healed endpoint → `/v1 ops 401` (the per-instance key lives only in `settings.endpoint` via `EndpointConverter`, never in the Room Endpoint row). Fresh onboard flows the key correctly; the L3 cold-start key-restore addresses keyless mDNS endpoints, but a full existing-install key-restore is still **owed**.
- Dead SSE path **CLOSED** (addaacd0): `observeSseSearch` + the 3 SSE bluff tests serving the unserved `/v1/search` were removed. A server-side `/v1/search` SSE aggregator, if ever wanted, is a fresh feature, not a revival.

auth-provider search — Auth-Token (login session) not threaded onto dynamic /v1 requests (2026-06-14, P0-1)

Symptom: after the 5-layer search fix shipped (1.3.9), search returned real results for NO-AUTH providers (Internet Archive) but NOT for login-required RuTracker/Kinozal (the operator's original report). **Root cause:** the Go `/v1/{provider}/search` path needs TWO credentials — Lava-Auth (per-instance key, fixed in 1.3.9) AND Auth-Token (the provider login session, `{provider}:cookie:{session}`, parsed in `lava-api-go/internal/handlers/handlers.go:118`). `ApiBackedTrackerClient.withAuth()` attached only Lava-Auth, so authed providers reached the server anonymous (`Type:"none"`) → `empty/login`. **Fix:** new `ProviderSessionTokenHolder` seam (parallel to `ApiBaseUrlHolder`); `ProviderLoginViewModel` writes the session on login; `TrackerClientModule`'s factory reads it; `ApiBackedTrackerClient.withAuth()` attaches Auth-Token: `{trackerId}:cookie:{session}` IN ADDITION to Lava-Auth when present. Additive — no-auth providers attach nothing, unchanged. **Affected files:** `core/tracker/client/.../ApiBackedTrackerClient.kt`, `.../ProviderSessionTokenHolder.kt` (new), `.../di/TrackerClientModule.kt`, `feature/login/.../ProviderLoginViewModel.kt`. **Verification:** `ProviderSessionTokenEndToEndWiringTest` — real `holder`→`factory`→`withAuth` over `MockWebServer` with an auth-gated dispatcher; PRIMARY assertion `RecordedRequest.getHeader("Auth-Token")` present for authed / absent for no-auth. Bluff-Audit: drop the attach → `authProviderSession_isThreaded_ontoTheAuthTokenHeader` FAILED (Auth-

Token absent → 401); reverted → :core:tracker:client + :feature:login GREEN. **ON-DEVICE VERIFY WITH REAL RUTRACKER CREDENTIALS OWED** (subagent rate-limit) — shipped in 1.3.10 for tester verification. **Fix commit:** 3b1b6a14.

remote/cloud GoApi wrongly given the LOCAL api-app key (2026-06-14, P1-4)

Symptom (edge): `OnboardingViewModel.withLocalApiKeyIfMissing()` read the LOCAL on-device api-app key for ANY keyless Endpoint.GoApi — correct for the local api-app, but a remote/cloud GoApi would get the local key → 401. No tests covered it. **Fix:** gate the key-read on `String.isLocalHost()` (loopback / RFC-1918 / .local); remote hosts stay keyless. **Affected:** `feature/onboarding/.../OnboardingViewModel.kt` + test. Bluff-Audit: remove the `isLocalHost` guard → remote endpoint wrongly keyed → test FAILED; reverted → GREEN (6/6). On-device mDNS flow not regressed. **Fix commit:** b3cb6de2.

READ_API_KEY permission not variant-suffixed → debug+release collide (2026-06-14, P2-1)

Symptom (QA-fidelity/security): the signature permission `digital.vasic.lava.permission.READ_API_KEY` was a fixed literal (the authority was already variant-suffixed). A device with BOTH debug+release variants co-installed hit `INSTALL_FAILED_DUPLICATE_PERMISSION` / broken grant — which cost hours of test-VM debugging in the search cycle (it masked the real grant path). **Fix:** `manifest.permission.READ_API_KEY` placeholder per build type — release byte-identical `...permission.READ_API_KEY` (existing grants survive), debug `...permission.dev.READ_API_KEY`. Both manifests merge-verified; `ApiKeyProvider.attachInfoForTest` now reads the variant-aware `BuildConfig.API_KEY_PERMISSION`. **Affected:** `app/build.gradle.kts`, `api-app/build.gradle.kts`, both `AndroidManifest.xml`, `ApiKeyProvider.kt`. **Fix commit:** 4026756c. Analysis: `docs/issues/2026-06-14-readapikey-permission-variant-analysis.md`.

2026-06-23 — H1 search-401: AuthInterceptor overwrote per-install handoff key

Root cause: `AuthInterceptor` in `core/network/impl/` attached the build-time LAVA_AUTH UUID using `OkHttp's .header(fieldName, value)` (replace semantics). Interceptors fire last in the `OkHttp` chain, so it unconditionally overwrote the per-install handoff key that `ApiClient` had already placed on the request under the same `Lava-Auth` header name. The engine received the wrong (build-time) credential and returned HTTP 401, which the app surfaced as "Something went wrong" on every search.

Affected files: `core/network/impl/src/main/kotlin/lava/network/impl/AuthInterceptor.kt`

Fix: Added an only-if-absent guard — the interceptor attaches the build-time UUID only when the request does not already carry the Lava-Auth header, so the per-install handoff key survives to the wire unchanged.

Verification test/challenge: core/network/impl/.../
AuthInterceptorHandoffKeyTest.kt (JVM, 5 tests, falsifiability-proven:
reverting the guard caused
whenPreExistingHandoffKeyIsPresent_interceptorMustNotOverwrite to
fail with expected:<[HANDOFF-KEY-B64]> but was:<[<build-time-UUID-
B64>]>). Challenge45 (C45) exercises the full on-device search-auth flow end-to-end.

Fix commit: b58ef78b

Forensic anchor: docs/issues/2026-06-23-search-401-rootcause-deepening.md; incident surfaced during operator testing of every search returning "Something went wrong".

2026-06-23 — api-app-17-release stale-binary (§6.Z wrong-binary)

Root cause: The 1071 rebuild failed mid-package at
:app:uploadCrashlyticsMappingFileRelease (transient Crashlytics DNS error) before :api-app:assembleRelease finished. :api-app:clean was not run before the next attempt, so the leftover output file was named *-17-* (matching the new version name) but its embedded android:versionCode still declared 16 from the previous build. The firebase-distribute.sh filename-only picker matched the filename and distributed the stale binary; the api-app release channel shipped versionCode 16 as release 17.

Affected files: scripts/firebase-distribute.sh

Fix: (1) api-app rebuilt clean (./gradlew :api-app:clean :api-app:assembleRelease) and distributed as version 18 (corrective). (2) firebase-distribute.sh hardened with a _assert_apk_versioncode guard that aapt2-dumps the picked APK's manifest and fatals if its actual versionCode differs from APP_VERSION_CODE — the filename matches the name, this gate matches the bytes.

Verification test/challenge: tests/firebase/
test_assert_apk_versioncode.sh (hermetic; exercises the new aapt content-guard with a positive case and a stale-versionCode negative case).

Fix commit: b58ef78b

Forensic anchor: .lava-ci-evidence/sixth-law-incidents/2026-06-23-apiapp-17-release-stale-binary.json; discovered by the conductor's own post-distribute aapt sweep, not a user report.

2026-06-23 — ApiHttpException type regression (IOException vs IllegalStateException)

Root cause: The §6.AC telemetry commit (68b6e650) introduced ApiHttpException as a typed HTTP-error wrapper extending IOException. Four existing tests in core:tracker:client asserted the historical contract that HTTP errors from ApiBackedTrackerClient extend IllegalStateException. The IOException supertype broke those assertions, causing 4 test failures surfaced during the integration test run after the telemetry commit landed.

Affected files: core/tracker/client/src/main/kotlin/lava/tracker/client/ApiBackedTrackerClient.kt

Fix: Changed ApiHttpException to extend IllegalStateException instead of IOException, preserving the original HTTP-error contract while retaining the structured statusCode/requestUrl/httpMethod context fields needed for §6.AC telemetry enrichment.

Verification test/challenge: The 4 pre-existing core:tracker:client tests asserting the IllegalStateException HTTP-error contract; all 4 GREEN in the 860/0 full JVM suite attested at commit a6d8cbf7. Regression surfaced and fixed within the same session (caught by integration testing before distribute).

Fix commit: 68b6e650

Forensic anchor: Caught during the full ./gradlew testDebugUnitTest run after the §6.AC telemetry commit; not a user report.

2026-06-23 — ApiKeyClientTest compile break (§6.AC analytics-param addition)

Root cause: The §6.AC telemetry commit (68b6e650) added an analytics:AnalyticsTracker constructor parameter to ApiKeyClient. ApiKeyClientTest was not updated in the same commit, so it stopped compiling. The break was not caught by the session's targeted test runs (which built androidTest APKs, not :app JVM unit tests) and was only surfaced by the subsequent full-suite attestation run.

Affected files: app/src/test/kotlin/lava/vasic/lava/client/handoff/ApiKeyClientTest.kt

Fix: Passed a no-op AnalyticsTracker boundary stub to ApiKeyClient's constructor in the test. Telemetry is an outermost boundary per §6.AC; no SUT was mocked and no production code was touched. The test's 3 existing key-read assertions were byte-unchanged and remain GREEN.

Verification test/challenge: ApiKeyClientTest (3 tests); GREEN in the 860/0 full Android JVM suite attestation at commit a6d8cbf7 (.lava-ci-evidence/test-runs/2026-06-23-android-jvm-suite.md).

Fix commit: a6d8cbf7

Forensic anchor: Surfaced by the comprehensive full-suite attestation run that followed targeted C44/C45 on-device validation; not a user report.

2026-06-23 — firebase-distribute SIGPIPE on head-7 truncation (process discipline)

Root cause: A head -7 truncation inserted in the distribute loop to limit output caused SIGPIPE to kill the upstream distribute process before it could write the last-version-debug pointer file. The first client-debug distribute of version 1072 appeared to succeed visually (Firebase upload completed) but left the pointer unwritten; the subsequent --release-only stage saw a stale pointer and blocked on the §6.AA release-after-debug-confirm gate.

Affected files: scripts/firebase-distribute.sh (process discipline note; the head -7 was removed from the distribute loop output path).

Fix: Removed the head -7 truncation from the distribute pipeline so the process completes cleanly before the shell closes the write end of the pipe. Pointer files are now written atomically after the full distribute subprocess exits.

Verification test/challenge: The §6.Z apt content-guard (_assert_apk_versioncode) confirmed all 4 binaries of the 1072 cycle distributed correctly after the fix (. lava-ci-evidence/.../2026-06-23-C00-1072-PASS.json). No dedicated hermetic test added (the SIGPIPE is a shell process-discipline issue, not a logic branch in the script).

Fix commit: 5dd7ae06

Forensic anchor: Observed in distribute loop output during the 1072 all-4 distribute cycle; re-run of client-debug distribute recorded correct last-version-debug 1072 pointer.

BUG-2026-06-24-A — Search streaming: back-press hang + no client-side timeout

Date: 2026-06-24 **Surface:** feature/search_result — SearchResultViewModel **Crashlytics §6.O closure log:** . lava-ci-evidence/crashlytics-resolved/2026-06-24-search-timeout-cancel.md

Root cause (Bug 1 — no cancellation on back-press): observeStreamMultiSearch() starts an Orbit intent {} coroutine that blocks on sdk.streamMultiSearch(...).collect {}. Back-press dispatches a new intent {} (onBackPressed) that only posts a Back side effect — it does NOT cancel the in-flight coroutine. On a slow provider (e.g. yts), the user taps Back, the screen navigates, but the streaming coroutine stays alive until OkHttp's 30 s readTimeout fires. User sees a ~30 s hang during which Back appears unresponsive.

Root cause (Bug 2 — no client-side timeout): `observeStreamMultiSearch()` had no `withTimeout` guard. A slow-but-not-stalled provider holds the `SearchResultContent.Streaming` spinner for up to 30 s with no Retry affordance — the `handleStreamEnd()` → Error path only triggers once `collect {}` returns.

Root cause (Bug 3 — engine unbounded search handler + stale YTS mirrors): `lava-api-go/internal/handlers/v1/search.go` passed `c.Request.Context()` (no deadline) to `p.Search()`. YTS failover loop: 4 mirrors × `perAttemptTimeout(8 s)` = up to 32 s, exceeding OkHttp's 30 s `readTimeout` on the client — direct source of the `SocketTimeoutException` in Crashlytics issue 9d4ad2f4.... Independently, the YTS mirror list led with `yts.mx` which is NXDOMAIN (confirmed 2026-06-24), forcing the client to burn all failover slots before timing out. Field evidence: Crashlytics key `provider=yts`, exception at `Http2Stream.takeHeaders` (read-headers hang, not a write/connect failure). Investigation doc: `docs/issues/2026-06-24-search-timeout-and-interrupt-rootcause.md`.

Affected files:

- `feature/search_result/src/main/kotlin/lava/search/result/SearchResultViewModel.kt` (Bug 1 + Bug 2 — `cancel-on-back` + `withTimeout`)
- `lava-api-go/internal/handlers/v1/search.go` (Bug 3 — `context.WithTimeout(18 s)` deadline wrapping `p.Search()` at line 69)
- `lava-api-go/internal/provider/curated/yts/client.go` (Bug 3 — `DefaultBaseURLs` refreshed: dead `yts.mx` dropped, 5 live mirrors: `yts.bz`, `yts.lt`, `yts.am`, `yts.gg`, `movies-api.accel.li`)

Fix:

- **Bug 1 (cancel-on-back):** Added `@Volatile private var activeSearchJob: Job? = null`. Inside `observeStreamMultiSearch()` captured the coroutine `Job` via `currentCoroutineContext()[Job]` and stored in `activeSearchJob`. `onBackClick()` now cancels any prior `activeSearchJob` before posting `Back`.
- **Bug 2 (client timeout):** Wrapped `collect {}` in `withTimeout(SEARCH_TIMEOUT_MS)` (25 000 ms, 5 s below OkHttp's `readTimeout`); `TimeoutCancellationException` caught locally, marks still-SEARCHING providers as `StreamStatus.ERROR`, routes to `SearchResultContent.Error` with Retry button. Generic `Exception` catch with `rethrowIfCancellation()` guard handles connectivity-lost similarly. §6.AC telemetry in both catch blocks.
- **Bug 3 (engine deadline + mirror refresh):** `handlers/v1/search.go:69` wraps `p.Search()` in `context.WithTimeout(c.Request.Context(), 18*time.Second)` so the engine ALWAYS returns (results or fast error) before the client's 30 s deadline. `yts/client.go` `DefaultBaseURLs` refreshed to 5 live mirrors (dead `yts.mx` dropped). Defensive prior-job cancel added in `observeStreamMultiSearch()` for future double-invocation safety (cleanup 1aa42536).

Verification tests:

Client (Android): feature/search_result/src/test/kotlin/lava/search/result/SearchResultViewModelCancelTimeoutTest.kt — 3 tests using REAL SearchResultViewModel + REAL LavaTrackerSdk + REAL DefaultTrackerRegistry; only the outermost network boundary faked via CompletableDeferred-based clients (dispatcher-agnostic indefinite suspension):

1. back_press_while_streaming_cancels_inflight_search_and_emits_Back
2. slow_provider_exceeding_timeout_surfaces_Error_with_Retry_affordance
3. back_press_after_completed_search_preserves_Content_and_emits_Back

Engine (Go):

- lava-api-go/internal/provider/curated/yts/search_total_deadline_test.go — TestSearch_ExceedsDeadline_ReturnsContextDeadlineExceeded (slow mock server + 18 s ctx deadline → confirms the engine surfaces an error, not a hang) + TestSearch_WithinDeadline_ReturnsResults.
- lava-api-go/internal/provider/curated/yts/search_total_deadline_test.go — TestReproduction_StaleMirrorListFails (NewClientWithMirrors(["yts.mx"]).Search → provider error; fresh list → real results in <2 s).

Falsifiability confirmed (§6.N / Seventh Law clause 1):

- Removed activeSearchJob?.cancel() → Test 1 isStillCollecting stays true → FAIL
- Removed withTimeout(SEARCH_TIMEOUT_MS) → Test 2 times out waiting for Error → FAIL
- Reverted DefaultBaseURLs to yts.mx-only → TestReproduction_StaleMirrorListFails reproduces yts.mx: provider: unknown error → FAIL; fresh list → PASS (Bluff-Audit in commit body 0e81730b).

Fix commits:

- 20d98914 — client Bug 1 (cancel-on-back) + Bug 2 (25 s withTimeout) + engine Bug 3 (18 s handler deadline in handlers/v1/search.go:69)
- 0e81730b — YTS stale mirror refresh (DefaultBaseURLs + TestReproduction_StaleMirrorListFails)
- 1aa42536 — code-review cleanups (defensive prior-job cancel; mirror-count comment fix)
- 3c1fa159 — version bumps: lava-api-go 2.3.33, CHANGELOG / snapshots for 1.3.11-1073 / api-app 0.2.11-19

Forensic anchor: Firebase Crashlytics NON_FATAL

9d4ad2f4d1a8b8697b1506402e045b81 (client release app

1:815513478335:android:456475e2ef4039d8cfd20a),

SocketTimeoutException at Http2Stream.takeHeaders, keys

feature=search / operation=streamMultiSearch / provider=yts, device HUAWEI TXZ-W09 / Android 12. Operator report (2026-06-24): "search does not work — no results; can't go back or interrupt it." Coordination analysis: docs/

issues/2026-06-24-search-timeout-coordination-analysis.md. §6.O closure log: `.lava-ci-evidence/crashlytics-resolved/2026-06-24-search-socket-timeout.md`

Challenge Test: owed — C-SEARCH-CANCEL per §6.AE (tracked in docs/workable_items.db). Parallel-authored on-device Challenge is the §6.O clause-2 e2e gate.

2026-06-24 — P0 CredentialsKeyHolder locked FATAL (Crashlytics 58a1335272bc)

Root cause: `CredentialsEntryRepositoryImpl.observe()` called `keyProvider()` (which invokes `CredentialsKeyHolder.require()`) unconditionally inside a `Room Flow.map` operator. When the key holder is in the **locked** state (normal at fresh app start or after session expiry), `require()` throws `IllegalStateException("credentials key holder is locked ...")`. That exception escaped the map operator and propagated onto the **main-thread Looper** via the `ViewModel`'s combine collector, causing a FATAL crash. The user was actively submitting searches (`lava_search_submit: mummy`, then `lava_search_submit: prince`) when the DAO emitted a background update that triggered the observation chain. 5 events / 1 user in 1.3.10.

Affected files:

- `core/credentials/src/main/kotlin/lava/credentials/CredentialsEntryRepositoryImpl.kt` — `observe()` now wraps `keyProvider()` in `runCatching`; locked-ISE → `emptyList()` + §6.AC recordWarning; all other throwables re-thrown
- `core/credentials/src/test/kotlin/lava/credentials/CredentialsEntryRepositoryImplTest.kt` — new test `observe` emits empty list when key holder is locked – no crash on search path (line 119)

Fix: `runCatching { keyProvider() }` guards the key-acquisition step. `isLockedKeyHolderError` predicate (type `IllegalStateException` + message prefix "credentials key holder is locked") routes the locked state to `emptyList()`. All other exceptions propagate unchanged. The locked state is expected and normal; crashing on it was the bug.

Verification test/challenge:

- Unit: `CredentialsEntryRepositoryImplTest` — `observe` emits empty list when key holder is locked (falsifiable: revert guard → `IllegalStateException` escapes → test FAILS)
- Challenge: C47 Challenge47CredentialsLockedSearchTest — OWED (§6.O clause 2 gate)

Fix commit: uncommitted at log creation; ships in 1.3.11-1073 **Forensic anchor:** Crashlytics issue 58a1335272bc4ee06595bda6302a670a, 5 FATALs on Samsung Galaxy S23 Ultra / Android 16 during search, version 1.3.10-1067. **§6.O closure log:** `.lava-ci-evidence/crashlytics-resolved/2026-06-24-credentials-keyholder-locked.md`

2026-06-24 — P1 LazyColumn nested-scroll 2nd site recurring FATAL (Crashlytics c7c8cccad09f) — FIX OWED

Root cause: A LazyColumn (or LazyVerticalGrid) is rendered inside a parent Column/Box with Modifier.verticalScroll(rememberScrollState()), giving the lazy layout unbounded vertical height. Compose throws
IllegalStateException: Vertically scrollable component was measured with an infinity maximum height constraints... at composition time. This is the §6.Q forbidden antipattern. The first site (TrackerSelectorList) was fixed in 1.2.3 (.lava-ci-evidence/crashlytics-resolved/2026-05-05-tracker-settings-nested-scroll.md), but the Crashlytics issue still recurred through 1.3.10-1067 (2026-06-14), confirming a **second independent site** exists. The breadcrumb (screen_view{MainActivity} only) does not identify which sub-screen triggered the 1.3.10 event. A §6.Q structural scan across all composables written/modified since 1.2.5 is required to locate the second site.

Affected files: UNCONFIRMED — to be filled in when §6.Q scan completes and fix is applied.

Fix: OWED. Pattern: replace LazyColumn with Column for bounded lists, OR apply Modifier.heightIn(max = ...) to constrain the lazy layout's measurement height.

Verification test/challenge:

- §6.Q structural regression test — OWED (targeting the identified composable)
- Challenge Test driving navigation to the offending screen — OWED (§6.O clause 2 gate)

Fix commit: OWED — ships in 1.3.11-1073 (pending §6.Q scan in this cycle)

Forensic anchor: Crashlytics issue c7c8cccad09f..., FATAL, first seen 1.2.3, last seen 1.3.10 (2026-06-14), Samsung Galaxy S23 Ultra / Android 16. Prior fix for site-1 documented in 2026-05-05-tracker-settings-nested-scroll.md. **§6.O closure log:** .lava-ci-evidence/crashlytics-resolved/2026-06-24-lazycolumn-nested-scroll-2nd-site.md

2026-06-24 — P2 TopicPageDto MissingFieldException for Internet Archive crawl topics (Crashlytics 8cde0ac208b3)

STATUS: PARTIAL — not fully resolved in 1073 (honest §6.J self-audit correction).

Root cause (corrected after reading the FULL Crashlytics subtitle): the exception lists Fields [id, title, author, category, torrentData, commentsPage] are required ... but they were missing — kotlinx lists the MISSING fields, so the Internet Archive WP0-* crawl-topic /topic2 response omits **ALL SIX** of TopicPageDto's fields (an archived web page, not a torrent topic).

Note author/category/torrentData are nullable but have NO default, so kotlin still requires the keys present. (An earlier draft of this entry wrongly assumed only [commentsPage] was missing — that was based on a reproduction fixture that did not match the real payload.)

Affected files (the partial change that DID land):

- core/network/api/src/main/kotlin/lava/network/dto/topic/TopicPageDto.kt — commentsPage given default = TopicPageCommentsDto(page = 1, pages = 1, posts = emptyList()) (line 13)
- core/network/api/src/test/kotlin/lava/network/dto/topic/TopicPageDtoSerializationTest.kt — new test file (2 tests, for the commentsPage-only case)

What 1073 does (necessary, NOT sufficient): the commentsPage default removes ONE field from the required set. The other five (id, title, author, category, torrentData) remain required, so the real IA crawl-topic payload **still throws** (now listing 5). The 1073 change therefore does not, by itself, stop the reported crash; it is retained as a valid partial improvement.

Why not a rushed blanket fix: defaulting id/title to "" + author/category/torrentData to null would stop the throw but risks masking real malformed-topic bugs for normal rutracker/rutor topics (§6.J). The proper fix (lava-api-go populating the shape for crawl items, OR a dedicated sparse-topic DTO, OR not requesting /topic2 for IA crawl topics) is a tracked FOLLOW-UP.

Impact + decision: NON_FATAL, 1 event, niche (IA crawl topics only), not a regression — does NOT block 1073 (which fixes the P0 + P1 FATALs + the search timeout). Full fix ships in a later build.

Fix commit (partial): cb6f76b2 (commentsPage default only); honest correction in the follow-up commit **Forensic anchor:** Crashlytics issue 8cde0ac208b3..., NON_FATAL, 1 event on Genymobile Pixel 9 / Android 16, version 1.3.9-1066, topic WP0-20230122202907-crawl897. **§6.O closure log:** . lava-ci-evidence/crashlytics-resolved/2026-06-24-topicpagedto-missing-field.md

LVA-008 — C11/C06 nested-NavHost search_input teardown crash (Activity-scoped inner-host lifecycle)

Status: FIX LANDED in working tree — device-gate verification PENDING (see "Verification" below). **Type:** Bug · **Severity:** P1 · **Workable item:** LVA-008

Symptom (CONFIRMED on device, 2026-06-08): the app PROCESS crashes at MainActivity destroy with
java.lang.IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED' (top frame
androidx.lifecycle.LifecycleRegistryKt.checkLifecycleStateTransition, LifecycleRegistry.kt:92), on the inner search/search_input?
query={query}&... NavBackStackEntry. Reproduced by
Challenge11Archive0rgAnonymousSearchTest (archive.org anonymous search)
AND Challenge06DownloadTorrentFileTest (both deep-nav search →

search_input → search_result → topic). Forensics: `.lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json`.

Root cause (CONFIRMED): the bottom-nav graph is mounted as a NESTED NavHost from a parent NavBackStackEntry (addNestedNavigation in `app/.../navigation/MobileNavigation.kt`). The inner NavController's host LifecycleOwner defaults to that parent entry. At activity-destroy the parent entry is driven straight to DESTROYED; the inner controller's host-lifecycle observer then walks its OWN back stack moving every entry to DESTROYED — including a search_input entry left INITIALIZED (created by the `popBackStack()`; `openSearchResult()` pattern + nested `saveState/restoreState`, never composed to CREATED). LifecycleRegistry rejects INITIALIZED → DESTROYED, killing the process.

Candidate-fix selection (per incident JSON ranking): prior device-FALSIFIED candidates — nav-compose 2.9.1 → 2.9.8, LenientTeardownRule (uncatchable process death), move-search-to-outer-host, atomic `popUpTo` replace, and the NavTeardownGuard ON_STOP pruner (the stranded entry is NOT in the public `currentBackStack StateFlow`, so a public-API walk cannot reach it). The TOP-RANKED previously-UNTRIED candidate is the inner-NavHost lifecycle scoping (`next_hypotheses_untried[0]`: "Scope/lifecycle of the inner `rememberNestedNavController` vs the outer host at destroy"). That is what this fix implements.

Fix: bind the inner (nested) NavHost's host LifecycleOwner to the Activity's view-tree LifecycleOwner instead of the parent NavBackStackEntry. The Activity reaches CREATED during its own normal lifecycle (driving inner entries to at least CREATED before any teardown) and its destroy path runs the inner entries through the regular RESUMED → ... → CREATED → DESTROYED backward pass rather than the parent entry's abrupt collapse to DESTROYED. Only `LocalLifecycleOwner` is re-pointed — `LocalViewModelStoreOwner` + `LocalSavedStateRegistryOwner` remain the parent entry, so nested-graph ViewModels still clear when the nested destination leaves the back stack (no leak / no scope regression).

Affected files:

- `core/navigation/src/main/kotlin/lava/navigation/ui/NavigationHost.kt` — new `activityScopedLifecycle: Boolean = false` param; when true, wraps the NavHost in `CompositionLocalProvider(LocalLifecycleOwner provides <view-tree owner>); new private rememberActivityScopedLifecycleOwner()` using `LocalView.findViewTreeLifecycleOwner()` (same pattern already used in `core/ui/.../ModalBottomDialog.kt`).
- `core/navigation/src/main/kotlin/lava/navigation/ui/MobileNavigation.kt` — `NestedMobileNavigation` passes `activityScopedLifecycle = true` to its `NavigationHost` (the outer `MobileNavigation` host is unchanged → default false, preserving existing per-destination lifecycle scoping there).

Verification:

- **Compile (DONE, captured):** `./gradlew :core:navigation:compileDebugKotlin --max-workers=2 --no-daemon --offline → BUILD SUCCESSFUL in 37s` (only pre-existing

deprecation/context-receiver warnings). The added defaulted param keeps all existing NavigationHost callers source-compatible.

- **Regression tests = the device Challenges (device-gate PENDING):** the load-bearing regression tests are the existing on-device Challenge11ArchiveOrgAnonymousSearchTest + Challenge06DownloadTorrentFileTest, which crash at teardown BEFORE this fix and must pass AFTER it. **No JVM/Robolectric regression test is feasible:** this is an androidx LifecycleRegistry teardown-ORDERING crash at real Activity destroy with a nested NavHost + a stranded INITIALIZED entry; the incident JSON proves it is even uncatchable by any in-process JUnit TestRule (it is process death inside ActivityThread.performDestroyActivity). Device-gate execution on this macOS host is BLOCKED by §6.AH-debt (container/VM emulator path does not yet boot here; no host-direct fallback permitted). The fix is therefore landed + compiled but **NOT yet device-verified** — it MUST be run on the Genymotion / emulator gate-host (C06 + C11) before this entry's status flips to "device-verified" and before any distribute. Prior sibling candidates were device-FALSIFIED, so this remains a hypothesis-with-strong-mechanism until the gate run, NOT a confirmed fix.

Falsifiability rehearsal (§6.N/§6.J, device-gate): a deliberate non-crashing break that the device Challenge would catch — revert activityScopedLifecycle = true to false in NestedMobileNavigation (the production state this fix changes). Expected on-device result: C06 + C11 crash again at MainActivity destroy with the identical IllegalStateException: State must be at least 'CREATED' to be moved to 'DESTROYED' on the inner search_input entry (reverting reproduces the exact prior failure). Re-apply true; re-run; the teardown crash must not recur. (The rehearsal is device-gated for the same reason the fix is: the failure is a real-Activity teardown-ordering event with no JVM equivalent.)

Fix commit: on the agent worktree branch (see git log for SHA; this entry authored in the same commit as the fix per §6.T.4). **Forensic anchor / incident JSON (status updated):** .lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json